Comparative Analysis of Image Processing Techniques for Banana Ripeness Classification Using Support Vector Machines

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Assignment Documentation

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Table of Contents

Table of Contents **2**
Introduction **3**
1.1 Problem Background 3
1.2 Objectives/Aims 4
1.3 Motivation 4
Literature Review **6**
Literature Review 1 6
Literature Review 2 7
2.1 Main Concept / Algorithm 1 : colour Based Feature Extraction 9
2.1.1 Adaptive HSV Feature Extraction 9
2.1.2 CIE L*a*b* Feature Extraction 11
2.2 Main Concept / Algorithm 2 : Watershed-Based Segmentation 13
2.2.1 Marker-Controlled Watershed 13
2.2.2 Morphological Boundary Recovery 14
2.3 Main Concept / Algorithm 3 - Contrast Enhancement and Texture Analysis (CLAHE & Thresholding) 15
2.3.1 CLAHE Contrast Enhancement 15
2.3.2 Otsu Thresholding and Texture Feature Extraction 16
2.3.3 Feature Representation for Ripeness Assessment 17
2.4 Support Vector Machine Model 18
2.4.1 Support Vector Machine 18
2.4.2 Radial Basis Function Kernel 19
2.4.3 Feature Standardisation 20
2.5 Analysis of the existing algorithm (Comparison) 20
Methodology **26**
3.1 System flowchart 26
3.2 Description of dataset (if Applicable) 27
3.3 Applications of the algorithm(s) 29
3.3.1 Banana Fruit-Mask Generation (Pre-processing) 29
3.3.2 Colour Based Feature Extraction 30
Algorithm 1: Adaptive HSV and CIE L*a*b* 30
3.3.3 Watershed-Based Banana Foreground Segmentation 31
Algorithm 1: Marker-Controlled Watershed Segmentation 31
Algorithm 2: Morphological Boundary Recovery 33
3.3.4 Contrast Enhancement and Texture Analysis (CLAHE & Thresholding) 34
Algorithm 1: CLAHE, Otsu Thresholding and Texture Feature Extraction 35
Algorithm 2: SVM Training and Classification 36
Result & Discussion **38**
4.1 Experimental Results 38

4.1.1 Original RGB Baseline Results 38
4.1.2 Adaptive HSV and CIE L*a*b* Results 40
4.1.3 Watershed-Based Segmentation 44
Marker-Controlled Watershed 44
Morphological Boundary Recovery 47
Overall Comparison 49
4.1.4 Contrast Enhancement and Texture Analysis (CLAHE & Thresholding) 50
4.2 Discussion and Critical Interpretation 54
4.2.1 Overall Performance Comparison 55
4.2.2 Interpretation of the Adaptive Colour Method 55
4.2.3 Interpretation of the Watershed Method 56
4.2.4 Interpretation of the CLAHE and Texture Method 57
4.2.5 Class-Level Comparison 57
4.2.6 Selection of the Best Method 58
Conclusion **59**
5.1 Achievements 59
5.2 Limitations and Future Works 59
5.2.1 Dataset Limitations 59
5.2.2 Foreground-Segmentation Limitations 60
5.2.3 Feature-Extraction Limitations 60
5.2.4 Future Work 61
References **62**
Appendix A **67**
Appendix B **68**

Introduction

1.1 Problem Background

Post-harvest losses in the global agricultural industry remain a critical economic challenge, often exacerbated by subjective and inconsistent manual evaluation methods (Lv et al., 2024). In the context of banana production, inaccurate monitoring during the supply chain leads to premature spoilage and significant food waste. To mitigate these inefficiencies, the implementation of an automated Fruit Ripeness Assessment system has emerged as a reliable, non-destructive solution. Fruit ripeness assessment refers to the technological evaluation of a fruit's physiological developmental stage to determine its optimal harvest time, storage viability, and overall market readiness (Arunima et al., 2024). By transitioning from human sensory judgements to computer vision-based methodologies, the agricultural sector can achieve high-throughput monitoring, standardisation, and sustainable quality control.

The foundation of automated agricultural inspection lies in the analysis of Visual Surface Characteristics. These characteristics encompass the observable external features of a physical subject, such as colour, texture, shape, and geometric contours, which serve as primary indicators of its internal health and biochemical composition. In image processing, visual surface characteristics are mathematically represented through pixel intensity distributions and spatial frequencies, enabling algorithms to detect anomalies without penetrating the protective skin of the fruit. Leveraging these non-destructive visual cues provides a highly practical alternative to chemical testing, allowing for real-time condition assessment and automated sorting along rapid production lines (Kakani et al., 2020).

A core component of the proposed image processing pipeline is Maturity Classification, which is defined as the systematic categorisation of agricultural produce into distinct developmental phases, ranging from underripe to overripe. For bananas, the transition through these maturity stages is heavily correlated with the degradation of chlorophyll and the synthesis of carotenoids, resulting in a distinct visual shift from green to yellow (Watharkar et al., 2021). Accurate maturity classification is essential for supply chain logistics, as it dictates the appropriate handling procedures, environmental storage conditions, and transportation durations required to prevent premature rotting. Advanced processing techniques, such as colour space conversions and thresholding algorithms, facilitate this classification by establishing precise, objective parameters for each ripening stage (Nisa et al., 2023).

Beyond basic maturity stage identification, agricultural products must undergo rigorous Surface Quality Grading. This process involves evaluating the overall external condition, textural uniformity, and visual appeal of the produce to assign it a specific commercial class or grade, which directly influences its consumer acceptance and market value (Guo et al., 2020). Surface quality grading ensures that premium fruits are segregated from those exhibiting minor aesthetic imperfections or severe structural defects, thereby standardising the product inventory. By utilising machine vision technologies to analyse surface integrity, automated grading systems consistently outperform manual human inspection, which is frequently prone to physical fatigue and subjective biases.

1.2 Objectives/Aims

Objective 1: To implement three banana image-processing pipelines, Adaptive HSV and CIE L*a*b*, Marker-Controlled Watershed with morphological boundary recovery, and CLAHE–Otsu with texture extraction on the 1,200-image Banalyzer dataset by the end of the project.

Objective 2: To train and optimise an RBF-SVM classifier for each image-processing pipeline using the official 1,000-image training set and evaluate each classifier on the same balanced 200-image test set using accuracy, precision, recall, macro F1-score and confusion matrices.

Objective 3: To compare the three proposed pipelines with an Original RGB baseline and identify the best-performing method based on test accuracy, macro F1-score, class-level performance, number of misclassifications and feature dimensionality by the completion of the experimental evaluation.

1.3 Motivation

The increasing demand for high-quality agricultural products has intensified the need for reliable and efficient fruit quality inspection methods. As consumer expectations continue to rise, producers and retailers are under greater pressure to ensure that fruits reaching the market satisfy rigorous quality standards in terms of ripeness, appearance, and freshness. Conventional manual inspection remains labour-intensive, time-consuming, and susceptible to human subjectivity, resulting in inconsistent grading decisions and increased post-harvest losses. Consequently, there is a growing need for automated image processing solutions capable of providing objective, repeatable, and efficient fruit ripeness assessment.

The proposed fruit ripeness assessment system possesses significant commercialisation value for the agricultural and food industries. By automatically categorising fruits into unripe, ripe, and overripe classifications while simultaneously evaluating surface quality through blemish and defect analysis, the system can assist farmers, distributors, retailers, and food processing companies in making consistent grading decisions. Accurate ripeness assessment improves inventory management, supports appropriate storage and transportation planning, and reduces the distribution of poor-quality produce to consumers. Furthermore, automated inspection severely limits labour dependency and inspection time, thereby improving operational efficiency and lowering overall quality control costs for agricultural enterprises.

Beyond its commercial applications, the project offers profound social benefits by directly contributing to the reduction of global food waste. Fruits that are harvested, transported, or sold at inappropriate maturity stages often deteriorate rapidly and become unsuitable for consumption, leading to unnecessary economic losses and severe environmental impacts. An objective and non-destructive image processing system enables better identification of fruit maturity and surface quality prior to distribution, allowing stakeholders to optimise harvesting schedules and supply chain logistics. This intervention contributes to improved food availability, more efficient resource utilisation, and greater environmental sustainability throughout the agricultural sector.

Crucially, this project explicitly aligns with the United Nations Sustainable Development Goal 9 (SDG 9): Industry, Innovation and Infrastructure, as illustrated in Figure 1.1 (United Nations, 2015). SDG 9

promotes technological innovation and the adoption of modern industrial solutions to improve productivity, efficiency, and sustainable economic development. By applying digital image processing techniques to automate fruit quality inspection, the proposed system demonstrates how computer vision technologies can modernise traditional agricultural practices and support the necessary digital transformation of the agri-food industry.



Figure 1.1 SDG9 Industry Innovation and Infrastructure (United Nations, 2015)

Furthermore, the project supports the Malaysia MADANI framework (Figure 1.2), particularly encompassing the core principles of Innovation (Daya Cipta) and Sustainability (Kemampanan) (Jabatan Penerangan Malaysia, 2023). The application of image processing technology encourages national innovation by replacing subjective manual inspection with intelligent, data-driven quality assessment systems. Concurrently, the system promotes sustainability by reducing agricultural food waste, improving resource efficiency, and supporting responsible production and distribution. These tangible outcomes are highly consistent with Malaysia's aspiration to develop a progressive, technology-driven, and sustainable economy that ultimately benefits both the industry and the wider society.



Figure 1.2 Malaysia Madani (Jabatan Penerangan Malaysia, 2023)

Literature Review

Literature Review 1

Zhuang et al. developed an optical-imaging method for identifying four banana maturity stages. The banana foreground was first segmented using an RB chromatic map, Otsu thresholding, and morphological post-processing. Three 48 × 48 pixel regions were then selected from the stalk, middle, and tip of each banana.

Features used

The colour representation contained information from three colour spaces:

- **HSV**
    - **H (Hue):** represents the dominant peel colour, such as green, yellow, or brown.
    - **S (Saturation):** represents colour purity or intensity.
- **CIE L*a*b***
    - **a*:** represents the green-to-red colour axis. This is useful because banana peel changes from green to yellow and brown during ripening.
    - **b*:** represents the blue-to-yellow axis and therefore measures the development of yellow peel colouration.
- **CIE L*ch**
    - **c (Chroma):** represents colour strength or vividness.
    - **h (Hue angle):** represents perceptual hue in cylindrical CIELAB coordinates.

The study found particularly significant differences between maturity stages in HSV hue, L*a*b* a*, and L*ch hue angle. It also separately evaluated:

- UP-LBP texture features.
- HOG local-shape features.
- colour appearance features.

Accuracy before and after applying colour features

The study did not report an SVM trained directly on raw images without feature extraction. Therefore, there is no true “before feature extraction” accuracy. Instead, the colour features can be compared with the alternative texture and shape features:

Table 2.1: Comparison of applying Features

SVM feature representation	Test accuracy

HOG shape features	82.6%
UP-LBP texture features	93.4%
HSV + L*a*b* + L*ch colour features	**99.2%**

Compared with HOG, the colour representation improved SVM accuracy by **16.6 percentage points**. Compared with UP-LBP, it improved accuracy by **5.8 percentage points**. During repeated 10-fold cross-validation, the colour-feature SVM obtained an average recall of approximately **0.94 ± 0.04**; the final independent test-set accuracy was 99.2%.

These results support combining HSV with L*a*b*: HSV describes intuitive peel-colour changes, while L*a*b* represents perceptual green–red and blue–yellow differences. However, the study used carefully selected banana regions and relatively controlled images, so its 99.2% accuracy should not be treated as directly comparable to results from a diverse internet-image dataset. (Zhuang et al. (2019))

Conclusion of literature review

Zhuang et al. (2019) investigated colour, texture, and shape features for identifying four banana maturity stages. After segmenting the banana foreground, the authors extracted colour information from three 48 × 48-pixel regions located at the stalk, middle, and tip. Their colour representation included the H and S components of HSV, the a* and b* components of CIE L*a*b*, and the chroma and hue-angle components of CIE L*ch. The colour-feature SVM achieved an overall test accuracy of 99.2%, compared with 93.4% for UP-LBP texture features and 82.6% for HOG shape features. These results indicate that peel-colour changes provide stronger information for banana maturity classification than local texture or shape alone.

Literature Review 2

Pardede et al. compared RGB, HSV, HSL, and L*a*b* colour representations for fruit-ripeness classification using SVM. Unlike Zhuang et al., this was not a banana-specific study. Its dataset contained mangoes, tomatoes, oranges, and apples, with ripe and unripe classes, producing eight classification categories. The images were resized to 100 × 100 pixels before feature extraction.

For every colour space, the researchers calculated a histogram for each of its three channels. Each channel histogram contained 256 bins, and the three histograms were concatenated:

[256 x 3 channels =768 features}]

For HSV, these features represented:

- **Hue histogram:** frequency of different fruit colours.
- **Saturation histogram:** distribution of weak and strong colours.
- **Value histogram:** distribution of dark and bright pixels.

The histogram approach does not preserve the spatial position of individual pixels. Instead, it describes the overall colour distribution of the image. This provides some robustness to changes in fruit position, but background pixels can influence the representation when foreground segmentation is not applied.

The researchers evaluated linear, RBF, and polynomial SVM kernels. For the polynomial kernel, they also compared different polynomial degrees.

Accuracy before and after applying HSV features

Table 2.2: Comparison of applying HSV Features

SVM configuration	Best reported accuracy
Best linear-kernel configuration	64%
Best RBF-kernel configuration	12%
HSV features + degree-6 polynomial kernel	**76%**

The final HSV model achieved:

Table 2.3: Final Result of HSV

Metric	Result
Accuracy	**76%**
Precision	**80%**
Recall	**76%**
F1-score	**78%**

The HSV degree-6 polynomial model was 12 percentage points more accurate than the best reported linear-kernel model. Nevertheless, this should not be interpreted as an improvement caused exclusively by HSV because both the colour representation and SVM kernel were changed. A proper HSV ablation would compare RGB and HSV using the same train/test split, preprocessing steps, kernel, and hyperparameters. (Pardede et al. (2019))

Conclusion of Literature Review

Pardede et al. (2019) evaluated RGB, HSV, HSL, and L*a*b* colour features for fruit-ripeness classification using SVM. Each image was resized to 100 × 100 pixels, after which a 256-bin histogram was calculated for each colour channel. Concatenating the three channel histograms produced a 768-dimensional feature vector. Among the tested configurations, HSV features combined with a sixth-degree polynomial SVM produced the best performance, obtaining 76% accuracy, 80% precision, 76% recall, and a 78% F1-score. The result demonstrates the usefulness of HSV histograms for representing overall fruit-colour distributions, although the method was evaluated on mango, tomato, orange, and apple images rather than bananas.

2.1 Main Concept / Algorithm 1 : Colour Based Feature Extraction

2.1.1 Adaptive HSV

The Hue, Saturation and Value (HSV) colour space separates colour information from brightness. Hue represents the dominant colour, saturation indicates colour purity, and value describes brightness. This separation is useful for banana-ripeness classification because peel colour changes progressively from green to yellow, brown and dark brown during ripening and deterioration.

Previous research supports the use of HSV features for fruit-ripeness classification. Pardede et al. (2019) represented each HSV channel using a 256-bin histogram and classified fruit ripeness using an SVM. Their HSV features combined with a sixth-degree polynomial SVM achieved an accuracy of 76%. Zhuang et al. (2019) also identified hue as a significant indicator of banana maturity. Their combined colour features and SVM classifier achieved 99.2% test accuracy, although this result included HSV, CIE L*a*b* and CIE L*C*h° features rather than HSV alone.

In this study, HSV features are calculated only from pixels identified as fruit by the mask generated in Part A. Instead of applying a constant saturation threshold to every image, an adaptive threshold is calculated from the saturation distribution of each banana:

$$T_S = \text{clip}\left(P_{10}(S_M), 0.04, 0.25\right)$$

Figure 2.1: Adaptive Saturation Threshold Calculation

where ‘P_10(S_M)’ is the 10th percentile of saturation values inside the fruit mask. The clipping operation prevents the threshold from becoming excessively low or high.

An informative-pixel mask is subsequently generated using the following condition:

$$M_I = M_F \cap \left[(S \geq T_S) \vee (C^* \geq T_C) \vee (L^* \leq 45)\right]$$

Figure 2.2: Informative Fruit-Pixel Mask Generation

where ‘M_F’ is the GrabCut fruit mask, ‘T_S’ is the adaptive saturation threshold, ‘T_C’ is the adaptive CIE chroma threshold, and L* <= 45 retains dark areas. The final condition is important for rotten bananas because severely damaged areas may have low saturation but still contain useful ripeness information. If the informative mask contains fewer than 50% of the fruit pixels, the complete fruit mask is used to avoid discarding too much of the banana.

The implemented HSV descriptor consists of:

- An 18-bin normalised hue histogram

- An 8-bin normalised saturation histogram
- An 8-bin normalised value histogram
- Statistical measurements of saturation and value, including mean, standard deviation, 10th percentile, 25th percentile, median, 75th percentile and 90th percentile
- Circular hue measurements
- Proportions of green, yellow, brown/orange and dark pixels

Hue is treated as a circular variable because its highest and lowest values represent neighbouring colours. It is transformed using:

$$\theta = \frac{2\pi H}{180}$$

Figure 2.3: Circular Hue-Angle Conversion

The mean sine, mean cosine and resultant length of 'theta' are then extracted. This prevents incorrect results when hue values occur close to the beginning and end of the OpenCV hue scale.

The adaptive approach allows the selected pixels to change according to the colour and illumination characteristics of each image. It is therefore intended to be more robust than a single fixed HSV threshold when the dataset contains different backgrounds, lighting conditions and banana-ripeness stages.

2.1.2 CIE L*a*b*

The CIE L*a*b* colour space was developed to represent colour in a perceptually more uniform manner. According to the CIE standard, L* represents lightness, a* represents the green-to-red axis, and b* represents the blue-to-yellow axis. The approximate interpretation is:

- L*=0: black
- L*=100: white
- Negative a*: green
- Positive a*: red
- Negative b*: blue
- Positive b*: yellow

These channels are relevant to banana-ripeness classification. Unripe bananas generally contain stronger negative a* values because of their green colour, while ripe bananas exhibit higher b* values because of yellow pigmentation. Overripe and rotten bananas typically contain lower L* values and increasing brown or dark regions.

In the implemented technique, OpenCV CIE L*a*b* values are converted to their standard approximate ranges:

$$L^* = \frac{L_{\text{OpenCV}}}{255} \times 100$$

$$a^* = a_{\text{OpenCV}} - 128$$

$$b^* = b_{\text{OpenCV}} - 128$$

*Figure 2.4: OpenCV-to-CIE L*a*b* Value Conversion*

Chroma is derived from the a* and b* channels:

$$C^* = \sqrt{a^{*2} + b^{*2}}$$

Figure 2.5: CIE Chroma Calculation

An adaptive chroma threshold is calculated for every image:

$$T_C = \text{clip}\left(P_{10}(C^*_M), 3, 18\right)$$

Figure 2.6: Adaptive Chroma Threshold Calculation

This threshold helps identify pixels with meaningful colour while adapting to differences between green, yellow and dark rotten bananas.

For each of the L*, a*, b* and C* channels, robust statistical features are extracted, including the mean, standard deviation, percentiles and median. Normalised histograms are also generated using ten bins for each L*, a* and b* channel.

Dark deterioration regions are detected using an image-specific lightness threshold:

$$T_{\text{dark}} = \text{clip}\left[\text{median}(L^*) - 1.5\,\text{IQR}(L^*), 12, 40\right]$$

Figure 2.7: Adaptive Dark-Pixel Threshold Calculation

The proportion of fruit pixels below this threshold is stored as an adaptive dark-spot feature. Consequently, a naturally dark rotten banana is evaluated relative to its own lightness distribution instead of using only one fixed threshold for all images.

Zhuang et al. (2019) found that a* was a significant attribute for distinguishing banana maturity stages. Similarly, Hou et al. (2015) used CIE L*a*b* colour characteristics with an RBF-SVM and reported

96.5% recognition accuracy for banana-ripeness classification. These findings indicate that CIE L*a*b* features can effectively represent the green, yellow and brown colour transitions associated with banana ripening.

In this study, CIE L*a*b* complements adaptive HSV: HSV represents intuitive colour and brightness distributions, whereas CIE L*a*b* provides lightness, opponent-colour and chroma measurements. Their combined features are supplied to the SVM classifier.

2.2 Main Concept / Algorithm 2 : Watershed-Based Segmentation

2.2.1 Marker-Controlled Watershed

Watershed segmentation is a region-based image segmentation method derived from mathematical morphology. It interprets a grayscale image as a topographical surface, where low-intensity regions represent catchment basins and high-intensity regions represent ridges. During a conceptual flooding process, neighbouring basins expand until they meet, forming watershed lines that represent region boundaries (Vincent & Soille, 1991).

A major limitation of conventional Watershed segmentation is **over-segmentation**. Small variations caused by noise, texture, shadows or surface irregularities may create many local minima, causing a single object to be divided into unnecessary regions. Marker-Controlled Watershed addresses this limitation by restricting the flooding process to selected foreground and background markers rather than allowing every local minimum to initialise a region.

Foreground markers can be generated using the **Euclidean Distance Transform (EDT)**. For a binary foreground object, EDT assigns each foreground pixel a value corresponding to its distance from the nearest background pixel. Pixels near the object boundary therefore have lower values, while pixels deeper inside the object have higher values. These internal regions can be used as reliable foreground markers before Watershed segmentation.

Ismail and Malik (2022) applied controlled marker-based Watershed segmentation in a real-time fruit grading system involving apples and bananas. Multiple fruits placed inside a lighting box were separated using Watershed segmentation together with EDT and connected-component analysis. The distance map was used to locate foreground peaks before assigning individual object regions, demonstrating the suitability of marker-based Watershed for fruit foreground separation.

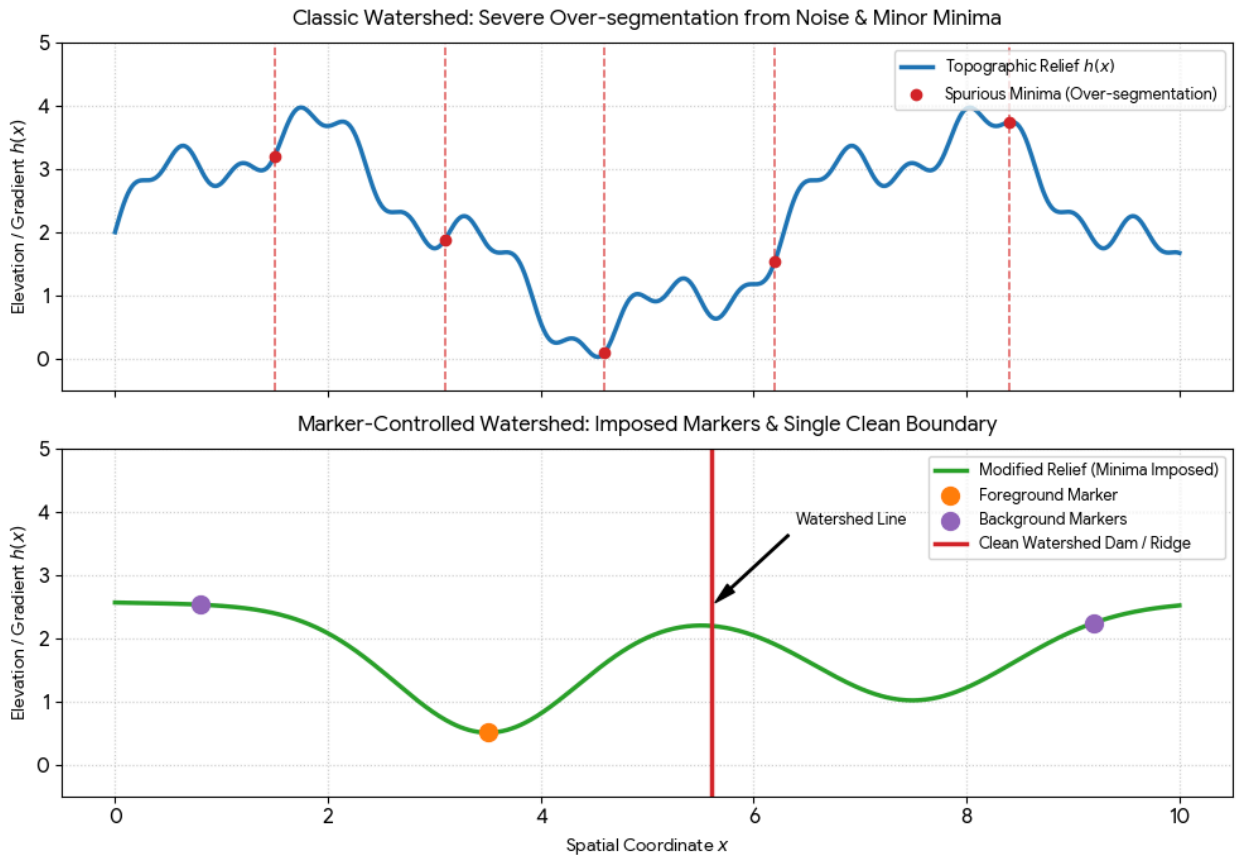


Figure 2.8: Conceptual Marker-Controlled Watershed Segmentation

The literature therefore indicates that Marker-Controlled Watershed is useful when object separation requires spatially controlled foreground and background regions. However, its effectiveness remains dependent on marker quality and the accuracy of the generated foreground boundary.

2.2.2 Morphological Boundary Recovery

Mathematical morphology modifies image regions according to their shape and spatial structure. Fundamental morphological operations include erosion, dilation, opening and closing. **Dilation** expands foreground regions, while **closing**, which consists of dilation followed by erosion, can close small gaps and improve the continuity of object regions (Haralick et al., 1987).

These operations are useful as post-segmentation refinement because an initial segmentation method may produce incomplete boundaries, small holes or disconnected foreground regions. Morphological processing can therefore refine a binary mask without requiring the entire segmentation procedure to be repeated.

The usefulness of morphological refinement has also been demonstrated in banana image analysis. Zhuang et al. (2019) first segmented banana foreground regions and subsequently applied morphological

processing to remove incorrectly segmented areas and fill holes before extracting colour, texture and shape information. This shows that morphological post-processing can improve the quality and continuity of a banana foreground mask before subsequent feature analysis.

For boundary recovery, morphological closing can reduce small discontinuities within a segmented object, while controlled dilation can restore pixels around an underestimated boundary. Nevertheless, excessive dilation can enlarge the foreground beyond the true object boundary. Morphological recovery must therefore be controlled to preserve relevant object regions without introducing excessive background information.

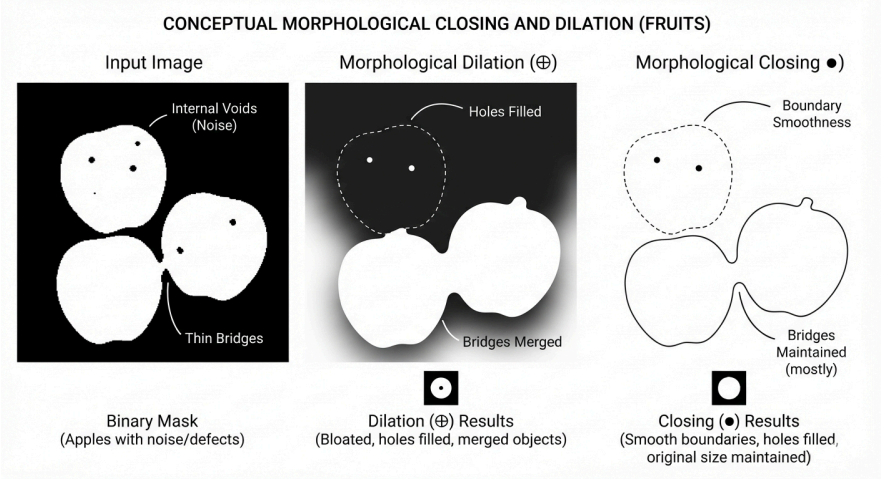


Figure 2.9: Conceptual Morphological Closing and Dilation

The reviewed literature identifies two complementary approaches for foreground segmentation and refinement. **Marker-Controlled Watershed** provides controlled region formation using foreground and background markers, while **Morphological Boundary Recovery** can refine incomplete masks through operations such as closing and dilation. Previous fruit and banana studies demonstrate that both Watershed segmentation and morphological post-processing are applicable to fruit foreground extraction before further image analysis (Ismail & Malik, 2022; Zhuang et al., 2019).

2.3 Main Concept / Algorithm 3 - Contrast Enhancement and Texture Analysis (CLAHE & Thresholding)

2.3.1 CLAHE Contrast Enhancement

Contrast enhancement is an important image-processing technique for improving the visibility of surface characteristics in banana images. Banana surfaces may contain uneven illumination, shadows and bright reflections, which can reduce the visibility of local intensity and texture variations. Contrast enhancement is therefore applied to improve the representation of these characteristics before feature extraction.

Contrast Limited Adaptive Histogram Equalization (CLAHE) enhances local contrast by dividing an image into small contextual regions and redistributing pixel intensities within each region (Zuiderveld, 1994). Unlike conventional global histogram equalisation, CLAHE limits excessive amplification through a clip limit, reducing the likelihood of strongly enhancing unwanted intensity variations.

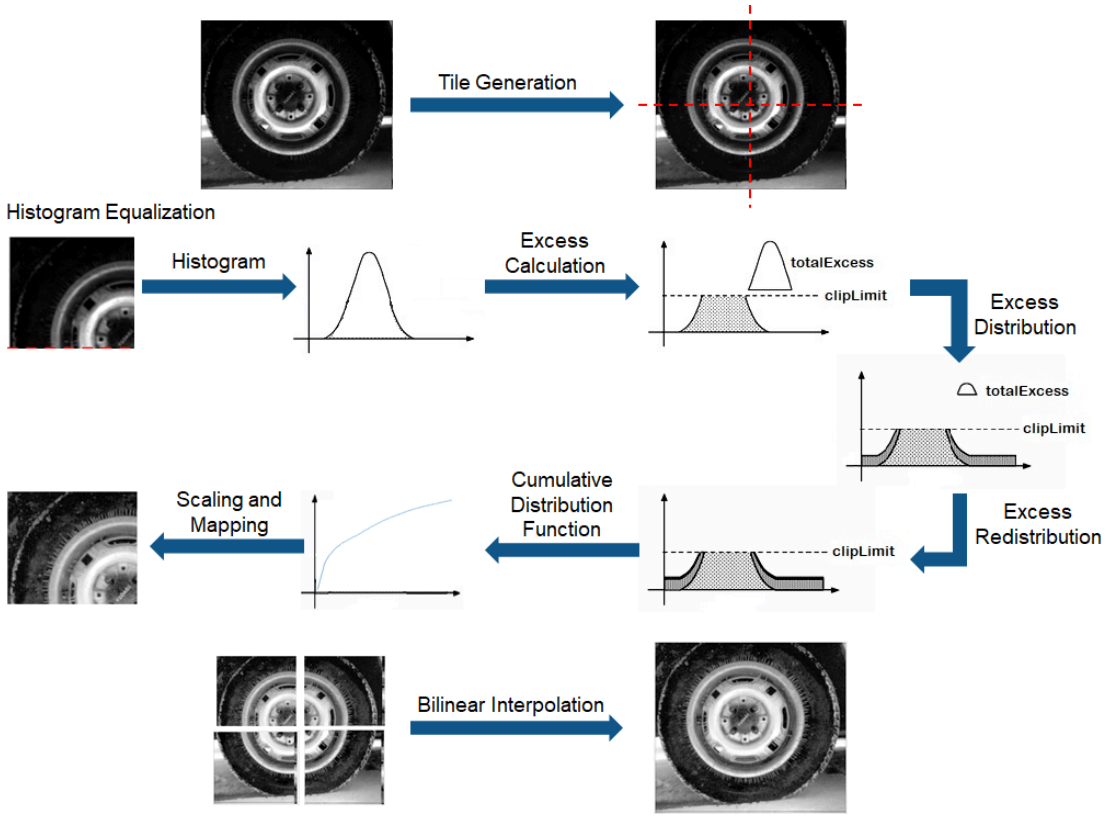


Figure 2.10: CLAHE Block Diagram / Flowchart (MathWorks, n.d.)

In the proposed implementation, the common Part A preprocessing is first applied to obtain the banana foreground image. The foreground is then converted from BGR to grayscale before CLAHE is applied using a clip limit of 2.0 and an 8 × 8 tile grid. The enhanced grayscale image is subsequently used for Otsu thresholding, texture extraction and grayscale statistical analysis.

CLAHE is suitable for Algorithm 3 because the proposed method focuses on local intensity and texture information rather than relying only on colour. Enhancing local contrast can make surface variations more distinguishable before the LBP, HOG and statistical feature-extraction stages.

2.3.2 Otsu Thresholding and Texture Feature Extraction

Following contrast enhancement, Otsu thresholding is applied to the CLAHE-enhanced grayscale image to generate a binary representation. Otsu's method automatically determines a threshold from the intensity distribution of the image, reducing the need for manual threshold selection (Otsu, 1979). The resulting binary image provides additional structural information for the feature representation.

The binary image is subsequently processed using morphological opening and closing with a 3 × 3 rectangular kernel. Opening helps reduce small isolated regions, while closing helps reduce small gaps and improve the continuity of the detected structures. The resulting cleaned binary image is used to calculate a binary ratio, representing the proportion of non-zero pixels in the thresholded image. The foreground ratio is calculated separately from the common foreground mask generated during Part A preprocessing.

Texture and structural information are then extracted from the CLAHE-enhanced grayscale image. Local Binary Pattern (LBP) is used to represent local texture patterns by analysing relationships between neighbouring pixels (Ojala et al., 2002). In the proposed implementation, LBP uses a radius of 2 pixels, 16 sampling points and the uniform pattern method. The resulting LBP values are represented as a normalised histogram.

$$\mathrm{LBP}_{P,R}(x_c, y_c) = \sum_{p=0}^{P-1} s(g_p - g_c)\, 2^p$$

Here, the s denotes the thresholding function. Where,

$$s(x) = \begin{cases} 1, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0 \end{cases}$$

- (x_c, y_c) is the **center pixel**,
- g_c is the **gray value of the center pixel**,
- g_p is the **gray value of the p-th neighbor** at radius R,
- P is the **number of sampling points** around the center.

Each comparison $s(g_p - g_c)$ is weighted by 2^p, forming a binary pattern that captures the local texture.

Figure 2.11: Local Binary Pattern (LBP) operator (Gunarathne, 2025)

Histogram of Oriented Gradients (HOG) is also used to represent local gradient and structural information. HOG describes the distribution of intensity-gradient directions within local image regions (Dalal & Triggs, 2005). The implementation uses 9 orientations, 16 × 16 pixels per cell and 2 × 2 cells per block with L2-Hys normalisation.

In addition, seven grayscale statistical features are extracted from the resized CLAHE-enhanced image. These consist of the mean, standard deviation, minimum, maximum, median, 25th percentile and 75th percentile. These statistics provide a compact representation of the overall intensity distribution of the enhanced image.

2.3.3 Feature Representation for Ripeness Assessment

The processed grayscale image is not used only as a visual output. Instead, the CLAHE-enhanced image and the Otsu-based binary representation are converted into numerical features for banana-condition classification.

The enhanced grayscale image is analysed using LBP and HOG. LBP describes local texture patterns by comparing neighbouring pixels with the centre pixel, allowing local surface variations to be represented statistically. HOG describes the distribution of intensity gradients and provides complementary information about local structures and directional patterns. In addition, seven grayscale statistical measurements are extracted to represent the overall intensity distribution of the enhanced image. The current implementation uses 16 sampling points with a radius of 2 for LBP, while HOG uses 9 orientations with 16 × 16 pixel cells and 2 × 2 cell blocks.

Two additional structural measurements are incorporated into the feature representation. The foreground ratio represents the proportion of pixels identified as foreground by the common Part A mask, while the binary ratio represents the proportion of non-zero pixels remaining after Otsu thresholding and morphological processing. These two measurements provide complementary information about the image structure.

The final feature vector is formed by concatenating the normalised LBP histogram, HOG features, seven grayscale statistical features, foreground ratio and binary ratio. With the current parameter settings, these components produce a 351-dimensional feature vector for each banana image. The resulting feature vectors are subsequently standardised using `StandardScaler` and supplied to the RBF-SVM for classification.

2.4 Support Vector Machine Model

2.4.1 Support Vector Machine

Support Vector Machine (SVM) is a supervised machine-learning algorithm used for classification. The main objective of SVM is to construct a decision boundary, known as a hyperplane, that separates observations belonging to different classes. The observations closest to the decision boundary are called support vectors. These observations are important because they determine the position and orientation of the boundary.

For a binary classification problem, the SVM decision function can be represented as:

$$f(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + b$$

Figure 2.12: SVM Decision Function

Where ‘x’ is the input feature vector, ‘w’ determines the orientation of the separating hyperplane, and b is the bias term. A new observation is classified according to the side of the decision boundary on which it lies.

SVM attempts to maximise the margin between classes. A larger margin generally allows the classifier to generalise better to unseen observations. However, image-derived features are rarely perfectly separable. Therefore, a soft-margin SVM introduces slack variables ‘ξi’ , which allow some training observations to be located inside the margin or classified incorrectly:

$$\min_{\mathbf{w},b,\xi} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n}\xi_i$$

Figure 2.13: Objective function of the soft-margin SVM optimization problem

subject to:

$$y_i\left(\mathbf{w}^T\phi(\mathbf{x}_i) + b\right) \geq 1 - \xi_i$$

Figure 2.14: Classification constraints with slack variables for the soft-margin SVM

The parameter ‘C’ controls the trade-off between producing a wide decision margin and penalising training errors. This soft-margin formulation enables SVM to handle overlapping classes and noisy observations, which are common in real-world banana images (Cortes & Vapnik, 1995).

2.4.2 Radial Basis Function Kernel

The banana-ripeness classifier in the present study uses the Radial Basis Function (RBF) kernel. A linear SVM can only construct straight decision boundaries in the original feature space. However, the colour distributions of unripe, ripe, overripe and rotten bananas may overlap and may not be separated effectively using a linear boundary.

The RBF kernel is expressed as:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\gamma\|\mathbf{x}_i - \mathbf{x}_j\|^2\right)$$

Figure 2.15: RBF kernel

where ‘x_i’ and ‘x_j’ are two feature vectors and ‘gamma’ determines the area of influence of an individual training observation. The kernel allows the SVM to construct nonlinear boundaries without explicitly calculating the coordinates of a high-dimensional transformed space. This process is commonly known as the kernel trick (Cortes & Vapnik, 1995).

The RBF kernel is appropriate for the present study because banana ripeness does not necessarily follow a simple linear change. For example, brown colour can indicate either an overripe or rotten banana, while yellow and green regions may occur simultaneously during transitional ripening stages. A nonlinear classifier can learn more complex relationships among the CIELAB colour features.

The RBF kernel contains two important parameters:

- C determines the penalty assigned to classification errors. A small 'C' produces stronger regularisation and a smoother decision boundary, while a large C attempts to classify the training observations more strictly and may increase the risk of overfitting.
- gamma determines the influence range of each training observation. A low value produces a smoother boundary with a wider influence range, whereas a high value produces a more local and complex boundary.

The implementation uses gamma="scale", which calculates gamma automatically as:

$$\gamma = \frac{1}{d \times \mathrm{Var}(X)}$$

Figure 2.16: Formula to Calculate Gamma

where 'd' is the number of input features and Var(X) is the feature variance. This setting adapts gamma to the scale and dimensionality of the training features (scikit-learn SVC documentation).

2.4.3 Feature Standardisation

Before the feature vectors are supplied to the SVM, StandardScaler standardises each feature according to:

$$z = \frac{x - \mu}{\sigma}$$

Figure 2.17: Formula of Standard Scaler

where 'x' is the original feature value, 'u' is its training mean and 'o' is its training standard deviation. Following standardisation, each feature has approximately zero mean and unit variance.

2.5 Analysis of the existing algorithm (Comparison)

Table 2.4 Comparison of Contemporary Approaches in Image Segmentation

Algorithm / System	**Example of Previous Work** ***(insert as citation)***	**Advantages**	**Limitations**	**Remarks**
Algorithm 1: GrabCut Foreground Segmentation	Rother et al. (2004)	Separates foreground and background using colour-distribution models and iterative graph-cut optimisation. It considers both pixel appearance and relationships between neighbouring pixels and can generate a relatively complete foreground mask.	Requires a suitable initial foreground estimate. Performance may decrease when foreground and background have similar colours or when the target touches the image boundary. Incorrect initialisation can remove valid object regions or retain unwanted background.	Used as the shared foreground-extraction stage before the three proposed image-processing techniques. It reduces background interference and provides a common fruit mask for subsequent processing.
Algorithm 2: Marker-Controlled Watershed Segmentation	Vincent and Soille (1991)	Controls Watershed region formation using predefined foreground, background and unknown markers, reducing the over-segmentation associated with conventional Watershed. Distance-based markers can identify reliable internal object regions and support the	Strongly dependent on marker quality. Conservative markers or weak boundaries may exclude valid object pixels, while inaccurate markers can generate incorrect regions. Watershed does not semantically recognise whether a region actually	Used as the primary segmentation stage of Pipeline 2. Euclidean Distance Transform and connected-component analysis generate the foreground markers before the final Watershed mask is produced.

		separation of neighbouring fruit regions.	belongs to a banana.	
Algorithm 3: Morphological Boundary Recovery	Haralick et al. (1987); Zhuang et al. (2019)	Morphological closing can reduce small gaps and improve mask continuity, while controlled dilation can recover foreground pixels around incomplete object boundaries. It is simple and effective as a post-segmentation refinement method.	Excessive closing or dilation can distort the original object shape or expand the foreground into unwanted regions. The result therefore depends on the selected structuring element and number of iterations.	Used as the second stage of Pipeline 2 after Marker-Controlled Watershed. It refines incomplete Watershed boundaries using closing and controlled dilation, while the recovered region is constrained by the shared fruit mask.
Algorithm 4: Otsu Global Thresholding with Morphological Refinement	Otsu (1979); Tripathi and Maktedar (2018)	Automatically determines a global threshold from the image intensity distribution and avoids manual threshold selection. Morphological opening and closing can remove small isolated regions and improve binary-mask continuity.	Performance decreases when foreground and background intensity distributions overlap or illumination is uneven. A single global threshold may remove relevant regions or retain unwanted areas.	Used in Pipeline 3 after CLAHE enhancement. Otsu thresholding generates the binary structural representation, while morphological processing refines the result before texture and structural features are extracted.

Table 2.5 Comparison of Contemporary Approaches in Contrast Enhancement

Algorithm / System	**Example of Previous Work *(insert as citation)***	**Advantages**	**Limitations**	**Remarks**
Algorithm 1: Global Histogram Equalization (GHE)	(Patel & Bhandari, 2019)	Simple to implement and computationally efficient. Improves the overall distribution of pixel intensities and increases global contrast.	Applies the same transformation to the entire image, so it may over-enhance noise and provide limited improvement in regions affected by uneven illumination.	Suitable for images with generally low global contrast, but less suitable for banana images containing shadows or different illumination levels across the fruit surface.
Algorithm 2: Contrast Limited Adaptive Histogram Equalisation (CLAHE)	(Reza, 2004)	Enhances local contrast by processing small image regions and limits histogram amplification to reduce excessive noise enhancement. It is effective for revealing local surface details under uneven lighting.	More computationally demanding than global histogram equalisation. Inappropriate parameter settings may produce excessive local enhancement or visible tile-related artefacts.	Most relevant to the proposed Algorithm 3. CLAHE is used to enhance local intensity variations before Otsu thresholding, LBP, HOG, and intensity feature extraction. Its local enhancement is suitable for banana images containing shadows, highlights, and surface defects.
Algorithm 3: Linear Contrast Stretching	(Ravikumar et al., 2021)	Simple and fast method that expands the image intensity range, improving global brightness and contrast while	Mainly provides global enhancement and may be sensitive to extreme intensity values. It does not	Useful as a lightweight enhancement method, but CLAHE is more appropriate for the proposed banana

(Min-Max Normalisation)		maintaining the basic intensity relationship between pixels.	explicitly enhance local regions affected by non-uniform illumination.	ripeness pipeline because local contrast enhancement is required before texture and threshold-based analysis.

Table 2.6: Comparison of Three Colour-Based Feature Extraction Approaches for Fruit-Ripeness Classification

Algorithm / System	**Example of Previous study**	**Advantages**	**Limitations**	**Remark**
Hybrid HSV, CIE L*a*b* and CIE L*C*h° features	Zhuang et al. (2019)	Combines complementary colour representations and achieved very high accuracy for four banana maturity stages. The features are relatively compact and interpretable.	Uses selected regions at fixed banana positions. Performance may decrease when banana position, orientation, background or illumination differs from the controlled image conditions.	Supports combining HSV and perceptually meaningful CIE colour features to represent banana-ripeness changes.

HSV histogram with polynomial SVM	Pardede et al. (2019)	Histograms provide a simple representation of the complete colour distribution and are independent of object orientation and exact pixel location.	Histograms discard spatial information, so two images with different colour arrangements may produce similar features. The dataset included mangoes, tomatoes, oranges and apples rather than bananas.	Supports the use of normalized HSV histograms, but indicates that HSV histograms alone may not be sufficient for high accuracy.
CIE L*a*b* colour features with RBF-SVM	Hou et al. (2015)	CIE L*a*b* separates lightness from green-red and blue-yellow colour components. This provides an interpretable representation of banana peel changes across seven ripening stages.	Individual L*, a* or b* values were insufficient when used alone. The method uses selected measurement regions and does not explicitly analyse dark spots, texture or complex backgrounds.	Supports using combined L*, a*, b* and derived chroma features with an RBF-SVM.

Methodology

3.1 System flowchart

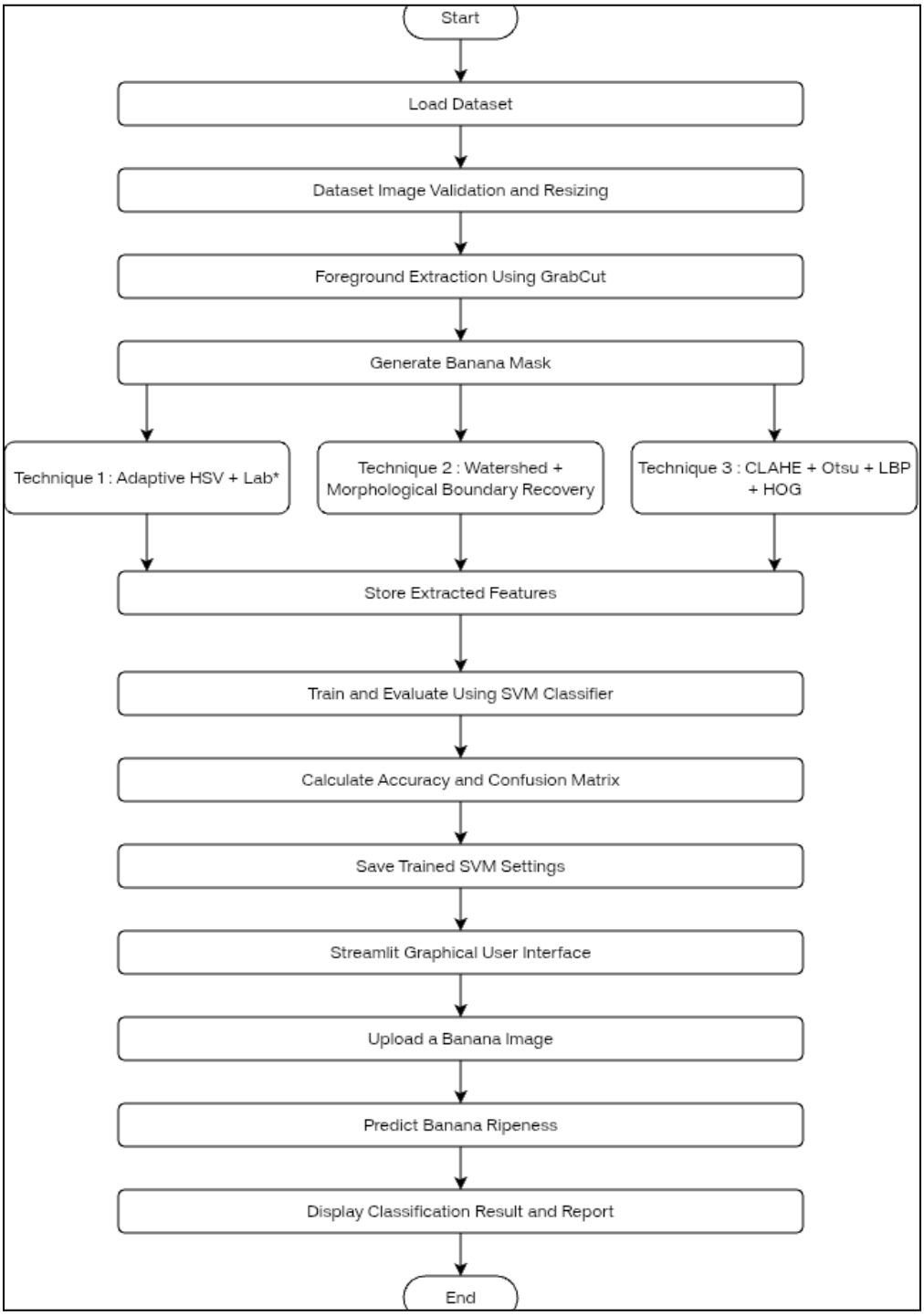


Figure 3.1: System Architecture and Process Flowchart

3.2 Description of dataset

The dataset used in this project is the Banalyzer – Banana Ripeness Classification Dataset, published by Charles C R on Kaggle. The dataset is intended for image classification and contains four banana-condition categories: unripe, ripe, overripe and rotten. The dataset is available from the official Kaggle dataset page.

Source:

R, C. C. (2025, November 24). *Banalyzer - Banana Ripeness Classification Dataset*. Kaggledatasets. https://www.kaggle.com/datasets/iamchaarles/banalyzer-banana-ripeness-classification-dataset

Direct inspection of the downloaded archive identified 1,200 valid RGB images, all stored in PNG format. The supplied dataset contains 1,000 training images and 200 test images. Each class contains 250 training images and 50 test images, resulting in a balanced distribution of 300 images per class. No unreadable image files were detected.

The raw images do not have a uniform resolution. A total of 172 distinct image dimensions were identified, although 809 images have a resolution of 1920 × 1080 pixels. This dimensional variation supports the need for letterbox resizing during preprocessing. The supplied dataset does not include a separate validation directory; therefore, a validation subset should be created only from the original training set, while the supplied test set remains unchanged for final evaluation.

Table 3.1: Distribution of Banana Images

Banana class	Training images	Test images	Total
Unripe	250	50	300
Ripe	250	50	300
Overripe	250	50	300
Rotten	250	50	300
Total	**1,000**	**200**	**1,200**

Dataset Structure

```
banana_data/
└── data/
    ├── train/
    │   ├── overripe/    250 images
    │   ├── ripe/        250 images
    │   ├── rotten/      250 images
    │   └── unripe/      250 images
    └── test/
        ├── overripe/     50 images
```

```
├── ripe/        50 images
├── rotten/      50 images
└── unripe/      50 images
```

Table 3.2: Dataset Field Table

Data field	Data type	Possible values	Description
split	Categorical string	train, test	Indicates whether the image belongs to the supplied training or test set.
class_label	Categorical string	unripe, ripe, overripe, rotten	Ground-truth condition obtained from the image's parent folder.
image	RGB image	PNG file	Contains the visual banana sample used for preprocessing and classification.
width	Integer	Variable	Original image width measured in pixels.
height	Integer	Variable	Original image height measured in pixels.
channel	Unsigned 8-bit integer	Red, green and blue	The three colour channels contained in each image.
pixel_value	Unsigned 8-bit integer	0–255	Intensity value for each pixel in each RGB channel.
filename	String	Varies by image	Unique filename used to identify the image within its class folder.

The Kaggle dataset card does not provide a detailed account of the original image-collection procedure, camera equipment or environmental controls. Consequently, claims about the capture conditions or original ownership of individual images should not be made without further information from the dataset creator.

Sample Image



Figure 3.2: Example of a banana image labelled as ripe in the Banalyzer dataset

3.3 Applications of the algorithm(s)

3.3.1 Banana Fruit-Mask Generation (Pre-processing)

Input: Banana image ‘I’, target size 256 x 256, GrabCut iterations ‘G=6’
Output: Resized image ‘I_R’, binary fruit mask ‘M_F’, foreground image ‘I_F’

```
1:  procedure GENERATE_FRUIT_MASK(I, G)
2:      if I cannot be decoded or does not have three channels then
3:          return FAILURE
4:      end if
5:
6:      IR ← Resize(I, 256, 256)
7:      R ← Define rectangle inside the image boundary
8:      MG ← Initialise GrabCut mask as probable background
9:      Mark image boundary in MG as background
10:     Mark central image region in MG as probable foreground
11:
12:     Apply GrabCut(IR, MG, R, iterations = G)
13:
14:     MF ← Pixels labelled foreground or probable foreground in MG
15:     MF ← MorphologicalOpening(MF, elliptical kernel 5 × 5)
16:     MF ← MorphologicalClosing(MF, elliptical kernel 5 × 5)
17:     MF ← Remove small and irrelevant connected components
18:
19:     if NumberOfForegroundPixels(MF) < 64 then
20:         MF ← Apply dark-background recovery strategy(IR)
21:     end if
22:
23:     if NumberOfForegroundPixels(MF) < 64 then
24:         return FAILURE
25:     end if
26:
27:     IF ← ApplyMask(IR, MF)
28:     return IR, MF, IF
29: end procedure
```

Input Example:



Figure 3.3 Example Input Image for Image Preprocessing

Output Example:



Figure 3.4 Example Output Image for Image Preprocessing

3.3.2 Colour Based Feature Extraction

Algorithm 1: Adaptive HSV and CIE L*a*b*

Input: Resized image 'I_R', fruit mask 'M_F'
Output: Feature vector 'X' containing 121 features

```
1:  procedure EXTRACT_ADAPTIVE_FEATURES(IR, MF)
2:      IHSV ← Convert IR from BGR to HSV
3:      ILAB ← Convert IR from BGR to CIE L*a*b*
4:
5:      H ← Hue channel of IHSV
6:      S ← Saturation channel of IHSV / 255
7:      V ← Value channel of IHSV / 255
8:
9:      L* ← (OpenCV L channel / 255) × 100
10:     a* ← OpenCV a channel − 128
11:     b* ← OpenCV b channel − 128
12:     C* ← SquareRoot(a*² + b*²)
13:
14:     Select H, S, V, L*, a*, b* and C* values where MF = 1
15:
16:     TS ← Clip(10thPercentile(S), 0.04, 0.25)
17:     TC ← Clip(10thPercentile(C*), 3, 18)
18:
19:     MI ← MF AND [(S ≥ TS) OR (C* ≥ TC) OR (L* ≤ 45)]
20:
21:     if NumberOfPixels(MI) < 0.5 × NumberOfPixels(MF) then
22:         MI ← MF
23:     end if
24:
25:     θ ← 2πH / 180
26:     Extract circular mean sine, mean cosine and resultant length
27:
28:     Extract statistical features from S, V, L*, a*, b* and C*
29:     Extract normalised histograms from H, S, V, L*, a* and b*
30:     Extract green, yellow, brown/orange and dark pixel
proportions
```

```
31:
32:     Tdark ← Clip[Median(L*) − 1.5 × IQR(L*), 12, 40]
33:     DarkSpotRatio ← Count(L* ≤ Tdark) / Count(MI)
34:
35:     Extract Laplacian texture features from L*
36:     X ← Concatenate all extracted features
37:
38:     return X
39: end procedure
```

Input Example:



*Figure 3.5 Example Input Image for Adaptive HSV and CIE L*a*b**

Output Example:

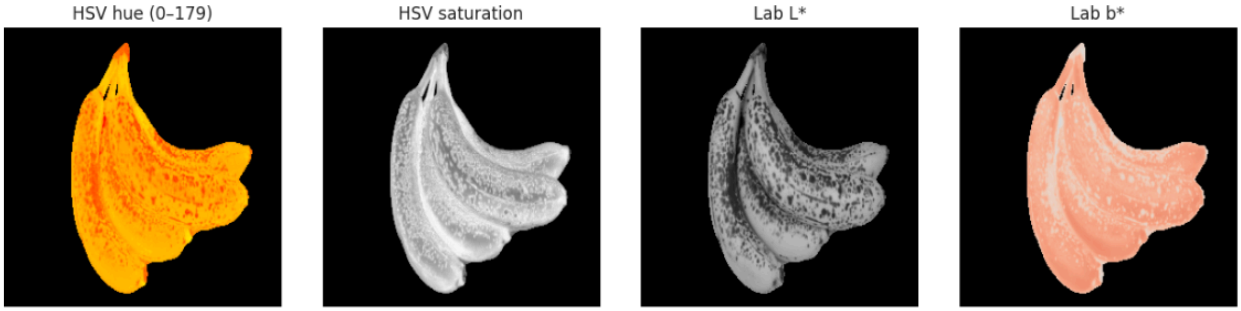


*Figure 3.6 Example OutputImage for Adaptive HSV and CIE L*a*b**

3.3.3 Watershed-Based Banana Foreground Segmentation

The Watershed-Based Segmentation method consists of two sequential algorithms: Marker-Controlled Watershed and Morphological Boundary Recovery. The shared preprocessing stage is first applied to obtain the resized image, shared fruit mask and foreground image. Marker-Controlled Watershed generates the initial banana mask, while Morphological Boundary Recovery refines the mask before the resulting RGB colour cut-out is used for classification.

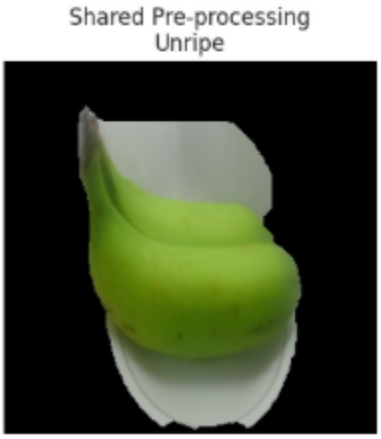
Algorithm 1: Marker-Controlled Watershed Segmentation

Input: Resized image I_R, shared fruit mask M_F, shared foreground image I_F
Output: Watershed mask M_W, Watershed colour cut-out IW

```
1: procedure MARKER_CONTROLLED_WATERSHED(I, MF, IF)
2:     MB ← Convert MF to binary mask
3:     D ← EuclideanDistanceTransform(MB)
4:     Dmax ← MaximumValue(D)
5:     MSF ← D ≥ (0.35 × Dmax)
6:     K ← CreateEllipticalKernel(3 × 3)
7:     ME ← Dilate(MB, K, iterations = 3)
8:     MU ← ME − MSF
9:     M ← ConnectedComponents(MSF, connectivity = 8)
10:    M ← M + 1
11:    Set markers in MU to 0
12:    L ← Watershed(I, M)
13:    MW ← Pixels where L > 1 AND MF > 0
14:    IW ← ApplyMask(IF, MW)
15:    Convert IW from BGR to RGB
16:    return MW, IW
17: end procedure
```

The shared fruit mask is converted into a distance map, and pixels reaching at least **35% of the maximum distance** form the sure foreground. A **3 × 3 elliptical kernel** and three dilation iterations define the surrounding unknown region. Connected components provide marker labels before Marker-Controlled Watershed generates the initial foreground mask.

Input Example:



Output Example:



3.7: Shared Pre-processing → Distance Transform → Sure Foreground → Markers → Watershed Mask → Colour Cut-Out

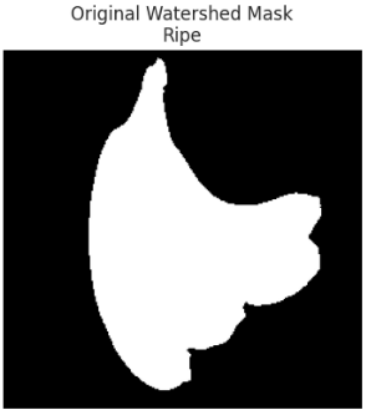
Algorithm 2: Morphological Boundary Recovery

Input: Watershed mask M_W, shared fruit mask M_F, shared foreground image I_F
Output: Recovered mask M_R, recovered colour cut-out IR

```
1: procedure MORPHOLOGICAL_BOUNDARY_RECOVERY(MW, MF, IF)
2:     K ← CreateEllipticalKernel(5 × 5)
3:     MR ← MorphologicalClosing(MW, K, iterations = 2)
4:     MR ← Dilate(MR, K, iterations = 1)
5:     MR ← MR AND MF
6:     if ForegroundPixels(MR) < 64 then
7:         MR ← MW
8:     end if
9:     IR ← ApplyMask(IF, MR)
10:    Convert IR from BGR to RGB
11:    return MR, IR
12: end procedure
```

Boundary recovery uses a **5 × 5 elliptical kernel**, two closing iterations and one dilation iteration. Closing reduces small gaps in the Watershed mask, while controlled dilation recovers surrounding boundary pixels. The recovered mask is restricted to the shared fruit mask so that the refinement does not expand into unrelated background regions.

Input Image:



Output Image:



Figure 3.8: Original Watershed Mask → Recovered Mask → Watershed Cut-Out → Boundary-Recovered Cut-Out

After segmentation, the colour cut-out is converted using the shared RGB feature procedure into a **64 × 64 × 3 = 12,288-dimensional** feature vector. Separate feature matrices of **1,000 × 12,288** for training and **200 × 12,288** for testing are generated.

An independent RBF-SVM is trained for each Watershed experiment. The initial Watershed Grid Search selected **C = 50** and **gamma = 0.0001**. The same SVM parameters are retained for the Boundary Recovery experiment so that the effect of the morphological refinement can be compared without changing the classifier configuration.

3.3.4 Contrast Enhancement and Texture Analysis (CLAHE & Thresholding)

The proposed Algorithm 3 uses contrast enhancement, thresholding, morphological processing and handcrafted texture features to represent the visual characteristics of banana images. The common preprocessing from Part A is first applied to obtain the banana foreground. The extracted foreground is then converted into a grayscale image before Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied using a clip limit of 2.0 and an 8 × 8 tile grid to enhance local contrast.

Otsu thresholding is subsequently applied to the enhanced grayscale image to generate a binary representation. Morphological opening and closing using a 3 × 3 rectangular structuring element are then performed to reduce small regions and improve the continuity of the binary result. The cleaned binary image is used to calculate a binary ratio, while the common foreground mask is used to calculate the foreground ratio.

For texture analysis, Local Binary Pattern (LBP) is extracted using a radius of 2 and 16 sampling points, producing a normalised histogram that represents local texture patterns. Histogram of Oriented Gradients (HOG) is also extracted using 9 orientations, 16 × 16 pixels per cell and 2 × 2 cells per block to describe local gradient and structural information. In addition, seven grayscale statistical features are calculated, consisting of mean, standard deviation, minimum, maximum, median, 25th percentile and 75th percentile.

These feature groups are concatenated with the foreground ratio and binary ratio to form a fixed-length feature vector.

The resulting feature vectors are standardised using StandardScaler and classified using an RBF-kernel Support Vector Machine (SVM) with C = 10 and gamma = "scale". The trained SVM predicts the ripeness class of each test image as Unripe, Ripe, Overripe or Rotten.

Algorithm 1: CLAHE, Otsu Thresholding and Texture Feature Extraction

Input: RGB banana image I
Output: Fixed-length feature vector F

```
procedure EXTRACT_FEATURES(I)
    _, M, Fg ← preprocess_image(I)
    G ← ConvertToGray(Fg)

    Gc ← ApplyCLAHE(G, clipLimit = 2.0,
                    tileGridSize = 8 × 8)

    B, T ← ApplyOtsuThreshold(Gc)
    B ← MorphologicalOpening(B, 3 × 3 kernel)
    B ← MorphologicalClosing(B, 3 × 3 kernel)

    Rb ← Mean(B > 0)

    Gc ← Resize(Gc, 64 × 64)

    L ← ExtractLBP(Gc, radius = 2,
                   samplingPoints = 16,
                   method = uniform)
    LH ← NormaliseHistogram(L)

    H ← ExtractHOG(Gc, orientations = 9,
                   pixelsPerCell = 16 × 16,
                   cellsPerBlock = 2 × 2)

    S ← [Mean(Gc), Std(Gc), Min(Gc), Max(Gc),
         Median(Gc), Percentile25(Gc),
         Percentile75(Gc)]

    Rf ← Mean(M > 0)

    F ← Concatenate(LH, H, S, Rf, Rb)

    return F
```

```
34: end procedure
```

Algorithm 2: SVM Training and Classification

Input: Training feature vectors Xtrain, training labels ytrain, testing feature vectors Xtest, testing labels ytest
Output: Predicted labels ypred and evaluation results

```
1:  procedure TRAIN_AND_CLASSIFY(Xtrain, ytrain,
2:                               Xtest, ytest)
3:
4:      Xtrain' ← StandardScaler(Xtrain)
5:      Xtest' ← StandardScaler(Xtest)
6:
7:      M ← CreateSVM(kernel = RBF,
8:                    C = 10,
9:                    gamma = scale)
10:
11:     Start training timer
12:     M ← Train(M, Xtrain', ytrain)
13:     Stop training timer
14:
15:     ytrain_pred ← Predict(M, Xtrain')
16:
17:     Start prediction timer
18:     ypred ← Predict(M, Xtest')
19:     Stop prediction timer
20:
21:     Acctrain ← Accuracy(ytrain, ytrain_pred)
22:     Acctest ← Accuracy(ytest, ypred)
23:
24:     R ← ClassificationReport(ytest, ypred)
25:     CM ← ConfusionMatrix(ytest, ypred)
26:
27:     Nmisclassified ← Count(ytest ≠ ypred)
28:
29:     Record training time
30:     Record average prediction time
31:
32:     return M, ypred, Acctrain, Acctest,
33:            R, CM, Nmisclassified
34: end procedure
```

Sample Input:



Figure 3.9: Sample Input Image for CLAHE, Otsu Thresholding and Texture Feature Extraction

Sample Output:



Figure 3.10: Sample Output Images for CLAHE, Otsu Thresholding and Texture Feature Extraction

Result & Discussion

4.1 Experimental Results

4.1.1 Original RGB Baseline Results

The baseline experiment was conducted using the original RGB images without image segmentation, foreground extraction or adaptive colour-feature extraction. Consequently, both the banana and background information were included in the input features. The purpose of this experiment was to

establish a reference performance for evaluating the contribution of the proposed preprocessing and adaptive HSV–CIE L*a*b* technique.

The baseline SVM correctly classified 172 of the 200 test images and achieved a test accuracy of 83.00%. The macro-average F1-score was 83.01%. Because each ripeness class contained 50 test images, the evaluation set was balanced.

Table 4.1: Original Baseline Results

Class	Precision	Recall	F1-score	Correct predictions
Unripe	0.94	0.66	0.78	33/50
Ripe	0.67	0.86	0.75	43/50
Overripe	0.89	0.82	0.85	41/50
Rotten	0.91	1.00	0.95	50/50
Overall/Macro	**0.85**	**0.83**	**0.83**	**172/200**

The rotten class produced the best overall result, with precision, recall and F1-score values of 0.94. Of the 50 rotten images, 47 were classified correctly, while three were incorrectly classified as overripe. The high result indicates that the substantial dark and brown colour characteristics of rotten bananas could be identified even without foreground segmentation.

The overripe class also achieved good performance, with an F1-score of 0.89. Forty-four overripe images were classified correctly, while three were classified as ripe and another three as rotten. These errors are reasonable because overripe bananas share yellow characteristics with ripe bananas and dark deterioration features with rotten bananas.

The main weakness of the baseline model was the unripe class. Although its precision was high at 0.95, its recall was only 0.70. Fifteen of the 50 unripe images were incorrectly classified as ripe. This means that when the model predicted an image as unripe, the prediction was generally reliable; however, it failed to identify 30% of the actual unripe samples.

The ripe class obtained a recall of 0.92 but a lower precision of 0.72. This low precision was mainly caused by the 15 unripe images incorrectly predicted as ripe. Therefore, the baseline model tended to confuse unripe bananas with ripe bananas, possibly because of variations in lighting, pale-green peel colours and background information.

Overall, the 83.00% baseline accuracy demonstrates that the original RGB images contain sufficient colour information for basic ripeness classification. However, the confusion between unripe and ripe bananas indicates that raw RGB information alone does not consistently isolate subtle peel-colour differences. This result provides the reference point for determining whether fruit segmentation and adaptive HSV–CIE L*a*b* features improve classification performance.

```
Baseline Test Accuracy : 86.00%
Baseline Macro F1      : 86.01%

Classification Report:

              precision    recall  f1-score   support

      unripe       0.95      0.70      0.80        50
        ripe       0.72      0.92      0.81        50
    overripe       0.90      0.88      0.89        50
      rotten       0.94      0.94      0.94        50

    accuracy                           0.86       200
   macro avg       0.88      0.86      0.86       200
weighted avg       0.88      0.86      0.86       200
```

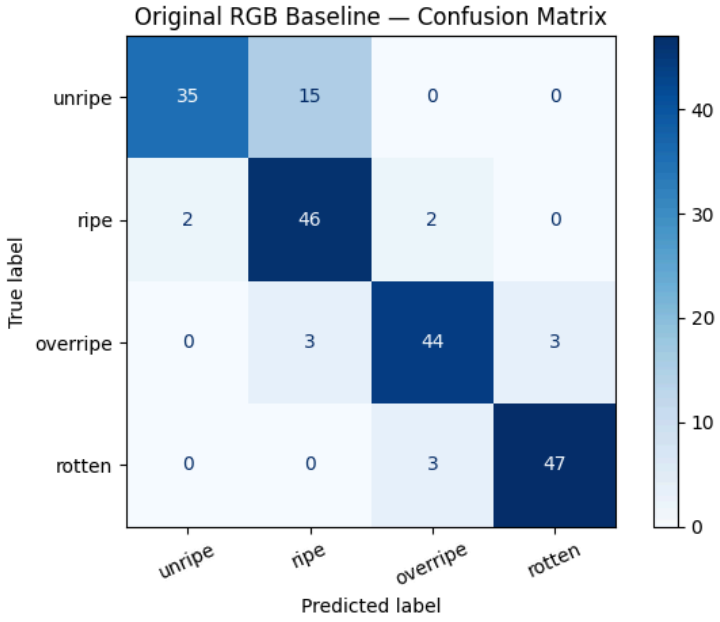


Figure 4.1: Confusion matrix of the baseline SVM classifier

4.1.2 **Adaptive HSV and CIE L*a*b* Results**

After applying fruit segmentation and the proposed adaptive HSV and CIE L*a*b* feature-extraction technique, the extracted feature vectors were standardised and classified using an RBF-SVM. Hyperparameter optimization selected C=100 and gamma=0.01. The best mean cross-validation accuracy was 92.99%.

The proposed model correctly classified 184 of the 200 test images, producing a test accuracy and balanced accuracy of 92.00%. Its macro-average precision, recall and F1-score were 91.91%, 92.00% and 91.89%, respectively.

*Table 4.2: Result of Adaptive HSV and CIE L*a*b**

Class	Precision	Recall	F1-score	Correct predictions
Overripe	0.8913	0.8200	0.8542	41/50
Ripe	0.9200	0.9200	0.9200	46/50
Rotten	0.9216	0.9400	0.9307	47/50
Unripe	0.9434	1.0000	0.9709	50/50
Overall/Macro	**0.9191**	**0.9200**	**0.9189**	**184/200**

The unripe class achieved the strongest result. All 50 unripe test images were classified correctly, resulting in a recall of 1.00 and an F1-score of 0.9709. This indicates that the adaptive colour features effectively represented the green characteristics of unripe bananas.

For the ripe class, 46 of the 50 images were correctly classified. Two ripe images were classified as overripe, while another two were classified as unripe. Nevertheless, the class achieved equal precision and recall values of 0.92.

The rotten class also obtained a high classification result. Forty-seven of the 50 rotten images were correctly identified, while three were classified as overripe. This produced a recall of 0.94 and an F1-score of 0.9307. The extracted dark-pixel, lightness and brown-colour features likely helped distinguish rotten bananas from the other classes.

The overripe class produced the lowest recall at 0.82. Of the 50 overripe images, 41 were correctly classified, four were predicted as ripe, four as rotten and one as unripe. These errors demonstrate that overripe bananas form a transitional category: some retain the yellow appearance of ripe bananas, whereas others contain dark spots similar to rotten bananas.

The cross-validation accuracy of 92.99% was close to the final test accuracy of 92.00%. The difference of only 0.99 percentage points suggests that the selected SVM configuration generalised consistently to the test set.

```
Best SVM parameters: {'svm__C': 100, 'svm__gamma': 0.01}
Best CV balanced accuracy: 0.9290
Test accuracy:          0.9200 (92.00%)
Test balanced accuracy: 0.9200
Test macro F1:          0.9189

Classification report:

              precision    recall  f1-score   support

    Overripe     0.8913    0.8200    0.8542        50
        Ripe     0.9200    0.9200    0.9200        50
      Rotten     0.9216    0.9400    0.9307        50
      Unripe     0.9434    1.0000    0.9709        50

    accuracy                         0.9200       200
   macro avg     0.9191    0.9200    0.9189       200
weighted avg     0.9191    0.9200    0.9189       200
```

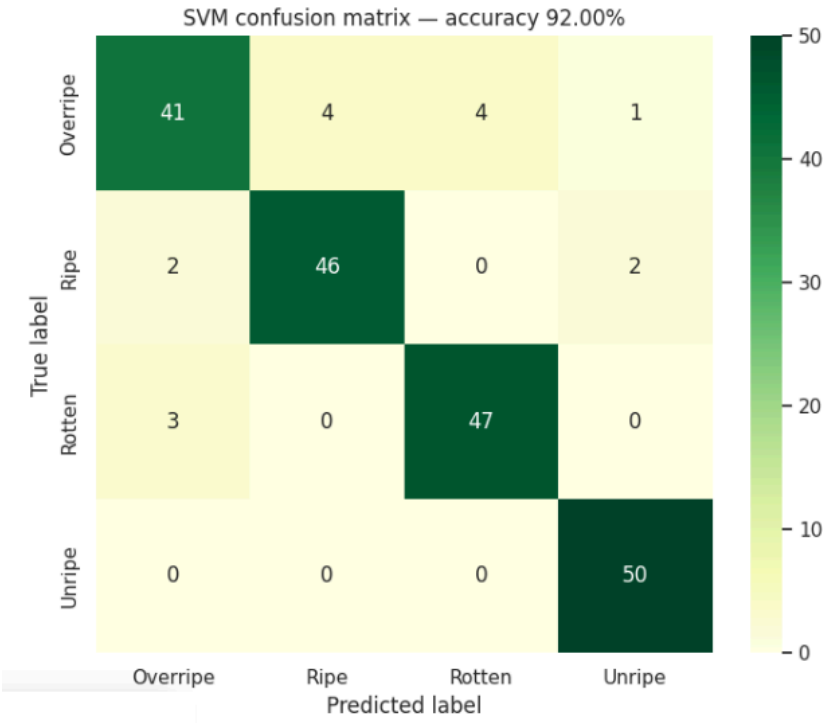


*Figure 4.2: Confusion matrix of the SVM classifier using the adaptive HSV and CIE L*a*b**

Comparison with the Original RGB Baseline

The performance of the proposed technique was compared with the original RGB baseline using the same 200-image test set.

*Table 4.3: Comparison Baseline VS Adaptive HSV CIE L*a*b**

Performance measure	**Original RGB baseline**	**Adaptive HSV CIE L*a*b***	**Difference**
Accuracy	83.00%	92.00%	**+6.00 percentage points**
Balanced accuracy	83.00%	92.00%	**+6.00 percentage points**
Macro precision	88.00%	91.91%	**+3.91 percentage points**
Macro recall	83.00%	92.00%	**+6.00 percentage points**
Macro F1-score	83.01%	91.89%	**+5.88 percentage points**
Correct predictions	167/200	184/200	**+12 images**
Incorrect predictions	33/200	16/200	**−12 images**

The proposed technique improved test accuracy from 83.00% to 92.00%, representing an improvement of six percentage points. The number of incorrect predictions decreased from 33 to 16, equivalent to a 51.51% reduction in classification errors:

Error reduction = (33-16)/33 x 100% = 51.51%

The most substantial improvement occurred in the unripe class. Its recall increased from 0.70 to 1.00, meaning the proposed model correctly identified all unripe test images. The precision of the ripe class also increased from 0.72 to 0.92 because fewer unripe images were incorrectly classified as ripe.

*Table 4.4: Comparison F1-Score Baseline VS Adaptive HSV and CIE L*a*b**

Class	Baseline F1-score	Proposed F1-score	Change
Unripe	0.80	0.9709	+0.1709
Ripe	0.81	0.9200	+0.1100
Overripe	0.89	0.8542	−0.0358
Rotten	0.94	0.9307	−0.0093

Although overall performance improved, the overripe F1-score decreased slightly. This suggests that the extracted features separated unripe and ripe bananas more effectively but still encountered difficulty defining the boundary between ripe, overripe and rotten stages.

Overall, the results demonstrate that foreground segmentation and adaptive HSV–CIE L*a*b* feature extraction produced a more informative representation than the original RGB baseline. The proposed technique increased accuracy, improved macro-level performance and substantially reduced confusion between unripe and ripe bananas. However, further improvement should focus on distinguishing overripe bananas from neighbouring ripeness stages.

4.1.3 Watershed-Based Segmentation

The Watershed-Based Segmentation experiment evaluates the initial **Marker-Controlled Watershed** and its subsequent **Morphological Boundary Recovery** refinement. Both methods use the same 1,000-image training set and 200-image held-out test set, containing 50 images for each of the four classes: unripe, ripe, overripe and rotten.

Marker-Controlled Watershed

Marker-Controlled Watershed was first applied after the shared preprocessing stage. The configuration used a foreground distance ratio of **0.35**, a **3 × 3** marker kernel and three foreground-envelope dilation iterations.

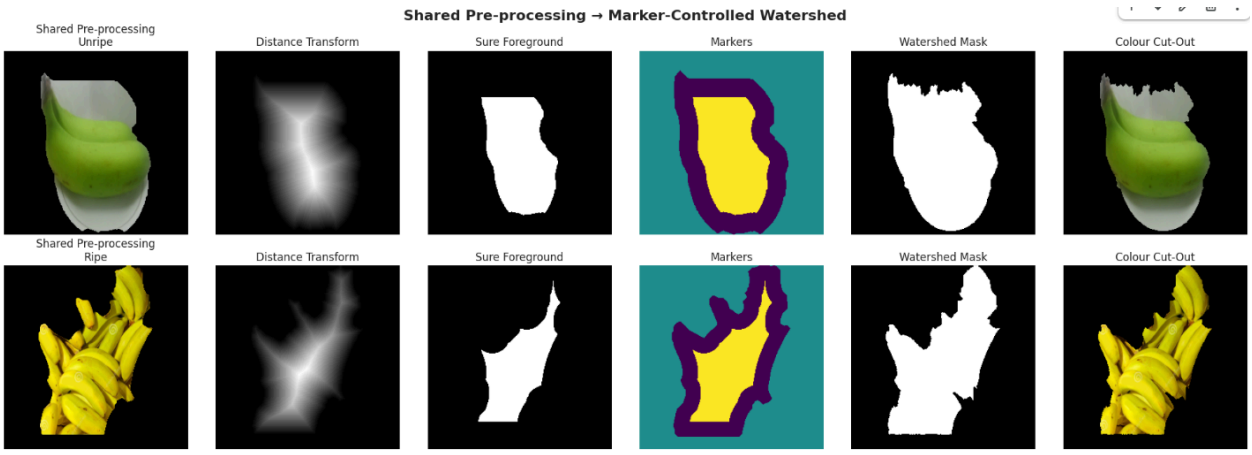


Figure 4.3: Marker-Controlled Watershed Processing Stages

Grid Search selected an RBF-SVM configuration of **C = 50** and **gamma = 0.0001**, with a training cross-validation accuracy of **82.40%**.

```
Watershed train: 100/1000
Watershed train: 200/1000
Watershed train: 300/1000
Watershed train: 400/1000
Watershed train: 500/1000
Watershed train: 600/1000
Watershed train: 700/1000
Watershed train: 800/1000
Watershed train: 900/1000
Watershed train: 1000/1000
Watershed test: 100/200
Watershed test: 200/200
Marker-Controlled Watershed SVM trained
Best C: 100
Best gamma: scale
Training CV accuracy: 82.40%
```

4.4: SVM Training Output showing C = 100, gamma = scale and CV accuracy = 82.40%

The Marker-Controlled Watershed model achieved **78.50% test accuracy** and **78.56% macro F1**, correctly classifying 157 of the 200 test images.

Table 4.5: Marker-Controlled Watershed Classification Performance

Class	Precision	Recall	F1-Score	Correct / Total
Unripe	0.78	0.70	0.74	35 / 50
Ripe	0.70	0.78	0.74	39 / 50
Overripe	0.83	0.78	0.80	39 / 50
Rotten	0.88	0.92	0.90	46 / 50
Macro / Overall	**0.80**	**0.79**	**0.79**	**157 / 200**

```
Watershed Test Accuracy : 79.50%
Watershed Macro F1      : 79.47%

Classification Report:

              precision    recall  f1-score   support

      Unripe       0.78      0.70      0.74        50
        Ripe       0.70      0.78      0.74        50
    Overripe       0.83      0.78      0.80        50
      Rotten       0.88      0.92      0.90        50

    accuracy                           0.80       200
   macro avg       0.80      0.79      0.79       200
weighted avg       0.80      0.80      0.79       200
```

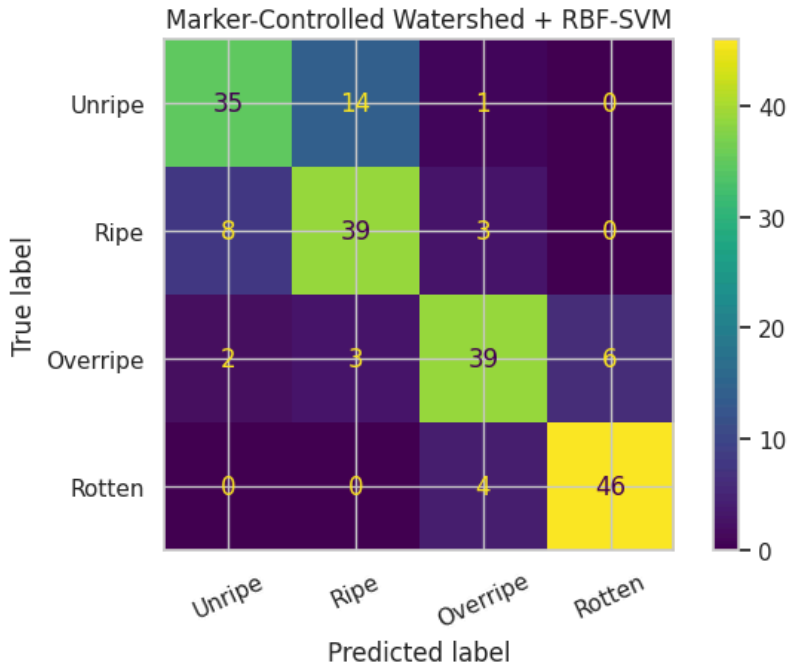


Figure 4.5: Marker-Controlled Watershed + RBF-SVM Confusion Matrix

The confusion matrix shows that the greatest difficulty occurred between neighbouring ripeness stages. Fourteen unripe images were classified as ripe, while eight ripe images were classified as unripe. Rotten obtained the strongest recall at **0.90**, with 45 of 50 samples correctly classified.

Compared with the Original RGB Baseline, Watershed accuracy decreased from **83.00% to 79.50%**, a difference of **−4.50 percentage points**. This suggests that foreground segmentation alone did not provide a better representation for the SVM and may have excluded some useful banana boundary or peel information.

Morphological Boundary Recovery

Morphological Boundary Recovery was subsequently applied to investigate whether refining the initial Watershed mask could reduce this information loss. A **5 × 5 elliptical kernel**, two closing iterations and one dilation iteration were used. The recovered mask was then constrained by the shared fruit mask.

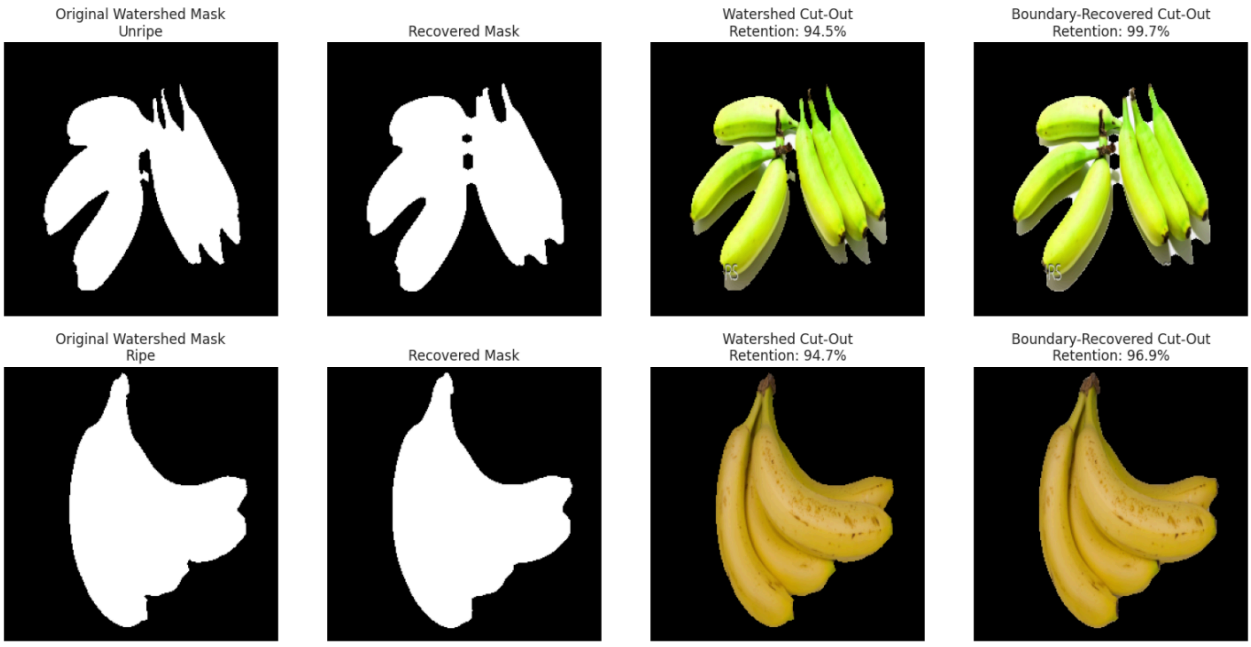


Figure 4.6: Original Watershed Mask versus Recovered Mask and Colour Cut-Out

The displayed examples show that boundary recovery retained a larger proportion of the shared fruit region. For the shown unripe image, retention increased from **94.5% to 99.7%**, while the shown ripe image increased from **94.7% to 96.9%**. These values refer only to the illustrated examples and are not overall dataset averages.

The recovered images produced training and testing feature matrices of **1,000 × 12,288** and **200 × 12,288**, respectively. A new independent RBF-SVM was trained using **C = 100** and **gamma = scale**, retaining the same classifier parameters as the original Watershed experiment.

The refined model achieved **80.00% test accuracy** and **80.04% macro F1**, correctly classifying 164 of the 200 test images.

Table 4.6: Watershed + Morphological Boundary Recovery Performance

Class	Precision	Recall	F1-Score	Correct / Total
Unripe	0.83	0.68	0.75	34 / 50
Ripe	0.68	0.78	0.73	39 / 50
Overripe	0.82	0.82	0.82	41 / 50
Rotten	0.88	0.92	0.90	46 / 50
Macro / Overall	**0.80**	**0.80**	**0.80**	**164 / 200**

```
Watershed + Recovery Test Accuracy : 80.00%
Watershed + Recovery Macro F1      : 79.95%

Classification Report:

              precision    recall  f1-score   support

      Unripe       0.83      0.68      0.75        50
        Ripe       0.68      0.78      0.73        50
    Overripe       0.82      0.82      0.82        50
      Rotten       0.88      0.92      0.90        50

    accuracy                           0.80       200
   macro avg       0.80      0.80      0.80       200
weighted avg       0.80      0.80      0.80       200
```

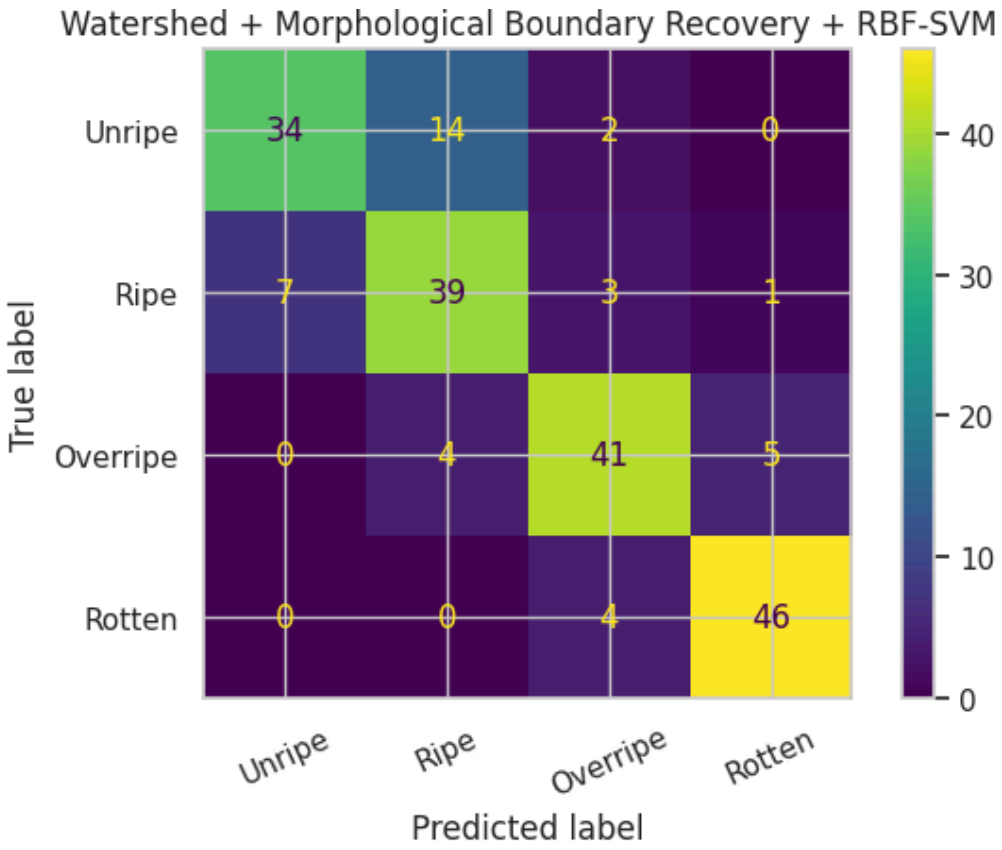


Figure 4.7: Watershed + Morphological Boundary Recovery + RBF-SVM Confusion Matrix

The strongest improvement occurred for the unripe class, where correct predictions increased from **35 to 46**. Overripe predictions also increased from 38 to 42. However, the refinement did not improve every class. Correct ripe predictions decreased from 39 to 34, while correct rotten predictions decreased from 45 to 42. In particular, 15 ripe images were classified as rotten, indicating that recovering additional foreground information also increased confusion between some classes.

Overall Comparison

Table 4.7: Comparison of Watershed-Based Segmentation Methods

Method	Test Accuracy (%)	Macro F1 (%)	Correct / Total
Original RGB Baseline	**83.00**	**83.00**	**167 / 200**
Marker-Controlled Watershed	79.50	78.56	157 / 200

Watershed + Morphological Boundary Recovery	**80.00**	**79.95**	**164 / 200**

Morphological Boundary Recovery improved the initial Watershed accuracy from **78.50% to 82.00%**, representing an increase of **3.50 percentage points**. Macro F1 similarly increased from 78.56% to 82.04%, while the number of correct predictions increased from 157 to 164.

Although the refined Watershed method remained **1.00 percentage points below the 83.00% Original RGB Baseline**, the improvement over the initial Watershed result indicates that mask refinement preserved additional information useful for classification. Therefore, the experiment demonstrates that the quality and completeness of the segmented foreground boundary can influence the effectiveness of Watershed-based banana ripeness classification.

	Method	Test Accuracy (%)	Macro F1 (%)	Correct / Total
0	Original RGB Baseline	83.5	83.44	167/200
1	Marker-Controlled Watershed	79.5	79.47	159/200
2	Watershed + Boundary Recovery	80.0	79.95	160/200

Figure 4.8: Final comparison chart — Original RGB Baseline vs Marker-Controlled Watershed vs Watershed + Morphological Boundary Recovery

4.1.4 Contrast Enhancement and Texture Analysis (CLAHE & Thresholding)

The proposed Contrast Enhancement and Texture Analysis pipeline was evaluated using the official test set of 200 banana images, consisting of 50 images from each of the four classes: Unripe, Ripe, Overripe and Rotten. The common preprocessing from Part A was first applied to extract the banana foreground. For Algorithm 3, the foreground image was converted to grayscale and enhanced using Contrast Limited Adaptive Histogram Equalization (CLAHE) with a clip limit of 2.0 and an 8 × 8 tile grid. Otsu thresholding followed by morphological opening and closing was then applied to obtain additional structural information.

The enhanced image was used to extract Local Binary Pattern (LBP), Histogram of Oriented Gradients (HOG) and grayscale statistical features. The extracted feature vector also included foreground ratio and the binary ratio obtained from the thresholded image. These features were concatenated and standardised using StandardScaler before being supplied to an RBF-kernel Support Vector Machine (SVM) with C = 10 and gamma = scale.

The proposed Hybrid feature representation contained **351 features**. The SVM achieved a **training accuracy of 99.30%** and a **test accuracy of 82.50%**, correctly classifying **165 out of 200 test images**. A total of **35 images were misclassified**. The SVM training time was **0.1989 seconds**, while the average prediction time was **0.000360 seconds per image**. These results indicate that the trained classifier can perform rapid prediction after feature extraction, although the difference between training and test accuracy suggests that the model does not generalise to the test data as strongly as its training performance indicates.

```
============================================================
Version B - Hybrid
============================================================
Training Accuracy : 99.30%
Testing Accuracy  : 82.50%
SVM Training Time : 0.1989 seconds
Average Prediction: 0.000360 sec/image
Misclassified     : 35
Support Vectors   : 709

Classification Report:
              precision    recall  f1-score   support

      Unripe     0.7826    0.7200    0.7500        50
        Ripe     0.7255    0.7400    0.7327        50
    Overripe     0.9149    0.8600    0.8866        50
      Rotten     0.8750    0.9800    0.9245        50

    accuracy                         0.8250       200
   macro avg     0.8245    0.8250    0.8234       200
weighted avg     0.8245    0.8250    0.8234       200
```

Figure 4.9: Overall Experimental Performance of the Hybrid CLAHE and Texture-Based SVM

The class-level results are presented in Table 4.8.

Table 4.8: Class-Level Classification Performance

Class	Precision	Recall	F1-Score	Support
Unripe	0.7826	0.7200	0.7500	50
Ripe	0.7255	0.7400	0.7327	50
Overripe	0.9149	0.8600	0.8866	50
Rotten	0.8750	0.9800	0.9245	50

Macro Average	0.8245	0.8250	0.8234	200
Weighted Average	0.8245	0.8250	0.8234	200

The results show that the Rotten class achieved the strongest performance, with a recall of 0.98 and an F1-score of 0.9245. This indicates that 49 of the 50 rotten banana images were correctly identified. The high recall suggests that the combination of contrast enhancement and texture-related features captured visual characteristics associated with severe deterioration effectively.

The Overripe class also produced relatively strong performance, with a precision of 0.9149, recall of 0.8600 and F1-score of 0.8866. This indicates that the extracted features were generally effective in recognising overripe bananas, although some samples were still confused with neighbouring ripeness conditions.

The Unripe class achieved a recall of 0.72 and an F1-score of 0.7500, while the Ripe class achieved a recall of 0.74 and an F1-score of 0.7327. These were the two weaker-performing classes. The lower scores indicate that the grayscale texture and structural representation did not always provide sufficient information to distinguish the subtle visual differences between earlier ripeness stages.

The difference between the training accuracy of 99.30% and test accuracy of 82.50% is 16.80 percentage points. This relatively large gap indicates that the trained SVM fitted the training feature vectors very well but experienced reduced generalisation on unseen test images. This may be associated with the limited number of training samples and the visual similarity between neighbouring banana-condition classes.

```
CLAHE Only Metrics:
      Class  Precision  Recall      F1
0    Unripe     0.6667    0.64  0.6531
1      Ripe     0.6207    0.72  0.6667
2  Overripe     0.9091    0.80  0.8511
3    Rotten     0.8600    0.86  0.8600

Hybrid Metrics:
      Class  Precision  Recall      F1
0    Unripe     0.7826    0.72  0.7500
1      Ripe     0.7255    0.74  0.7327
2  Overripe     0.9149    0.86  0.8866
3    Rotten     0.8750    0.98  0.9245
```

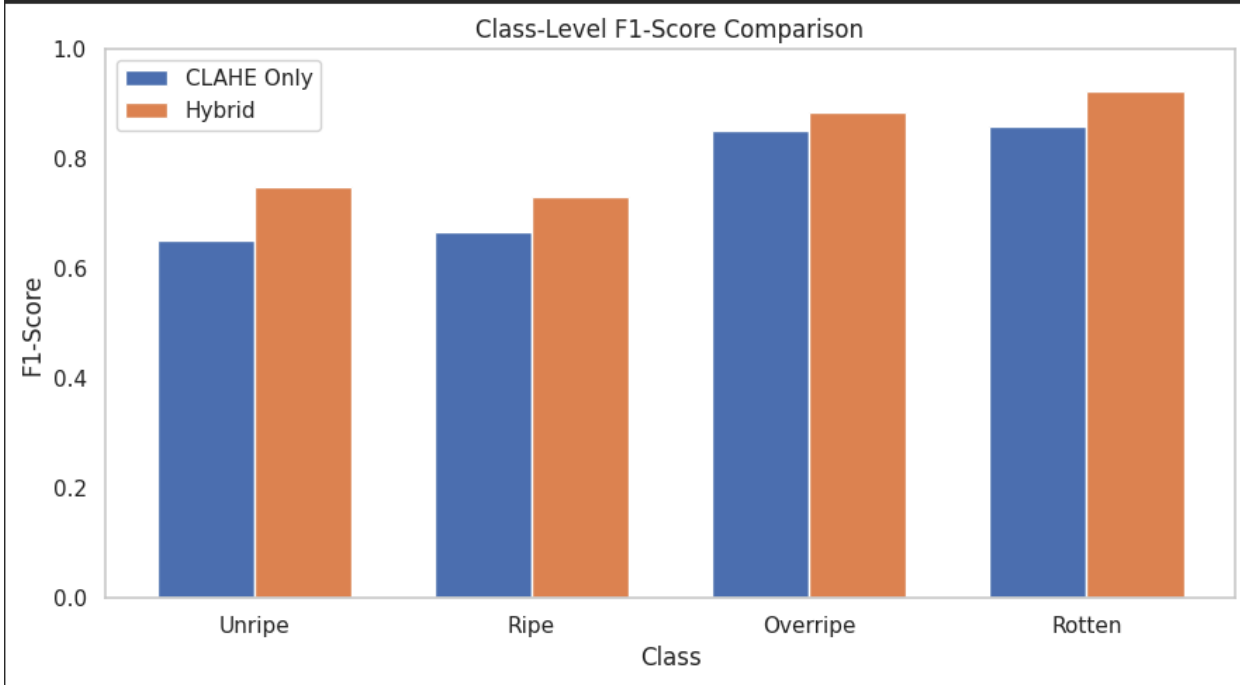


Figure 4.10: Classification Performance

Figure 4.10 presents the precision, recall and F1-score obtained for each banana-condition class. Rotten achieved the highest recall of 0.98 and the highest F1-score of 0.9245, while Ripe recorded the lowest F1-score of 0.7327. The results indicate that the proposed feature representation was more effective for identifying bananas with stronger deterioration characteristics than for distinguishing the visually similar unripe and ripe stages.

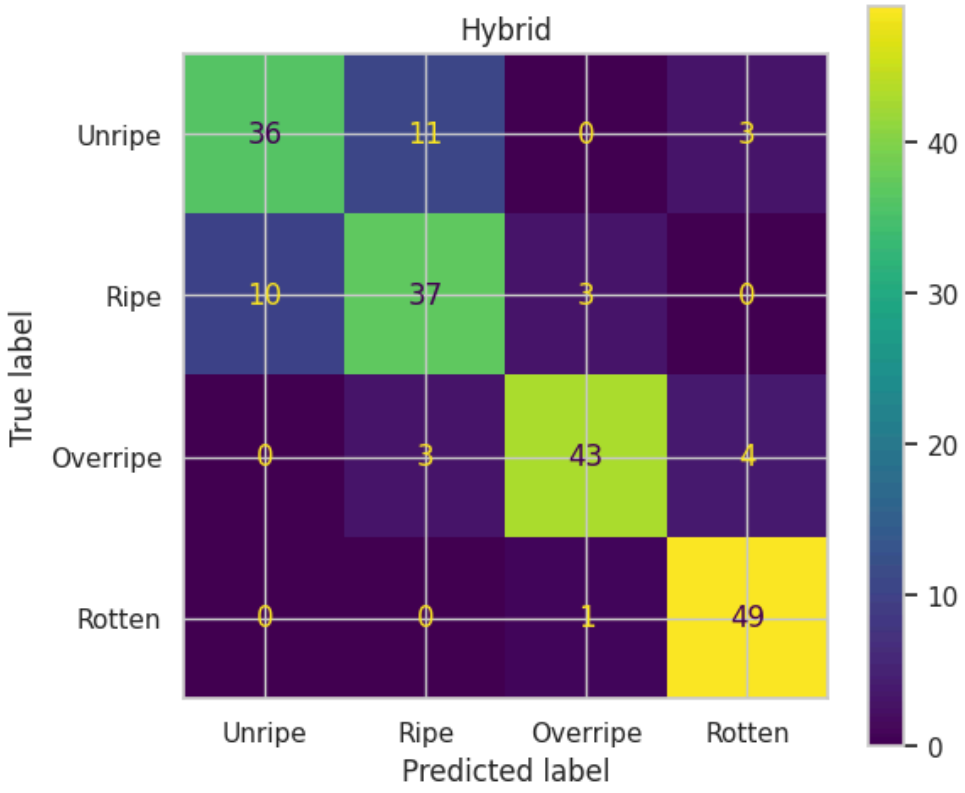


Figure 4.11: Confusion Matrix

Figure 4.11 presents the confusion matrix of the proposed Hybrid CLAHE and texture-based SVM. The model correctly classified 165 of the 200 test images, while 35 images were incorrectly classified. The confusion matrix shows the distribution of errors between the four banana-condition classes and provides a more detailed view of the classification behaviour than overall accuracy alone.

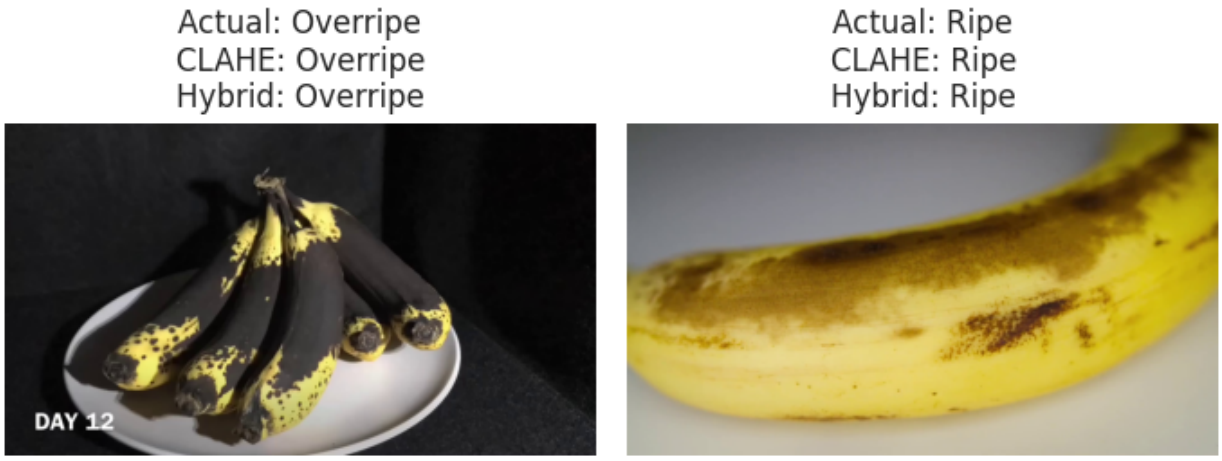


Figure 4.12: Sample Banana Ripeness Predictions

Figure 4.12 presents representative test images together with their actual and predicted classes. The examples demonstrate that the proposed pipeline can correctly classify banana images across all four condition categories.

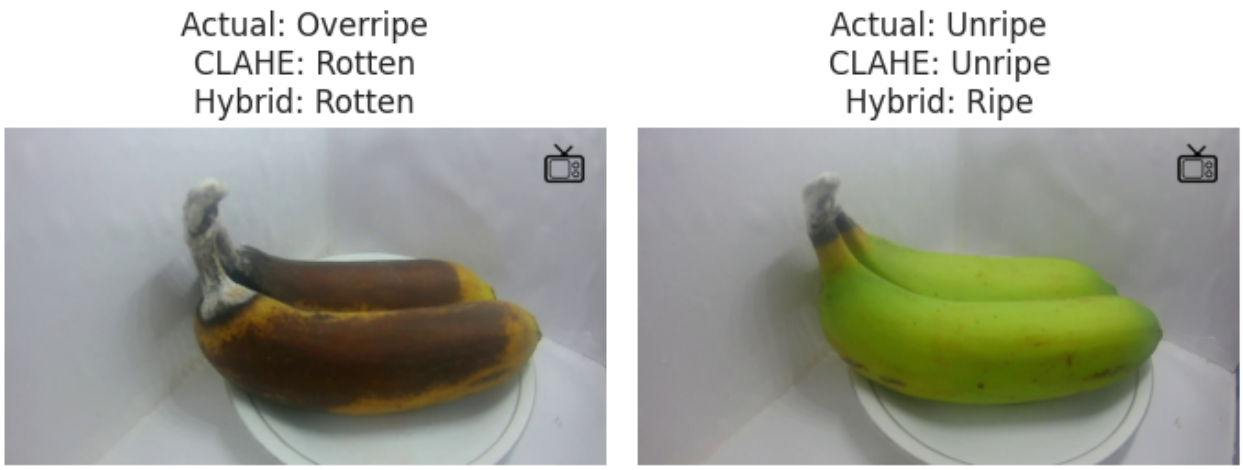


Figure 4.13: Misclassified Banana Images

Figure 4.13 presents representative examples of incorrectly classified banana images. These examples are useful for analysing the limitations of the proposed feature representation, particularly where neighbouring ripeness stages exhibit similar texture, intensity and structural characteristics.

4.1.5 Streamlit result

Streamlit: https://imageprocessing-9eu2mhevku9l56yxsngnjn.streamlit.app/

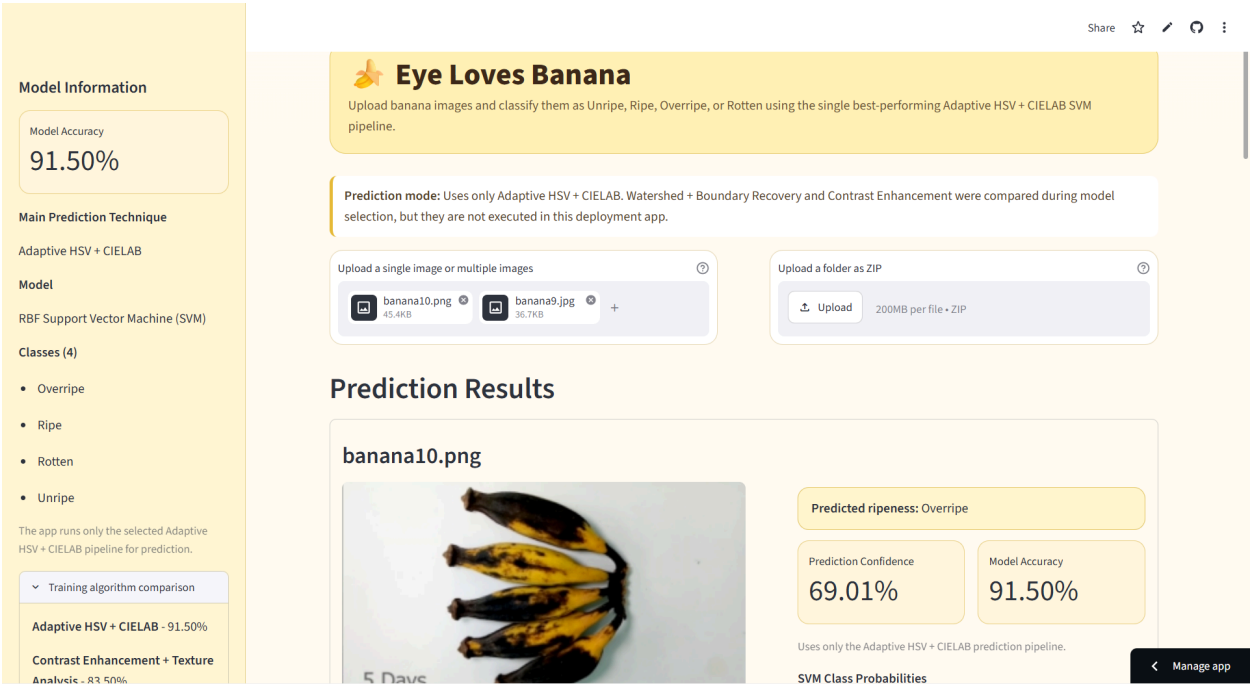


Figure 4.11: Streamlit-based deployment interface for banana ripeness classification

A Streamlit-based web application named Eye Loves Banana was developed to demonstrate the deployment of the proposed banana ripeness classification system. The application provides a simple user interface that allows users to upload a single banana image, multiple images, or a folder in ZIP format for prediction. As shown in Figure 4.11, the sidebar presents the deployed model information, including the model accuracy, main prediction technique, classifier type, and the four supported classes: Unripe, Ripe, Overripe, and Rotten.

The deployed application uses only the best-performing pipeline, namely the Adaptive HSV + CIE L*a*b* method combined with an RBF-SVM classifier. The other two methods, namely Watershed + Morphological Boundary Recovery and Contrast Enhancement + Texture Analysis, were compared during the experimental stage but were not selected for deployment because they achieved lower classification performance. Therefore, the Streamlit system applies only the selected adaptive colour pipeline during prediction.

In the main content area, the uploaded banana image is displayed together with the prediction result. For the example shown in Figure 4.11, the system predicted the uploaded banana image as Overripe with a prediction confidence of 69.01%. The interface also shows the overall model accuracy and a bar chart of SVM class probabilities, allowing the user to observe how strongly the classifier supports each ripeness category. This improves the interpretability of the prediction, as the user can see not only the final class label but also the relative confidence across the four classes.

In addition, the application includes a visual explanation section that shows how the uploaded image is processed through the selected pipeline. This section helps demonstrate the intermediate stages of the Adaptive HSV + CIE L*a*b* approach before the final SVM classification is produced. A reset button is also provided to clear the current upload session, while a PDF report download function enables users to save the prediction result in report form. Overall, the Streamlit application demonstrates that the trained model can be deployed in a practical and user-friendly manner for banana ripeness prediction.

4.2 Discussion and Critical Interpretation

The three proposed approaches produced different classification performances because they represent banana ripeness using different visual information. The Adaptive HSV and CIE L*a*b* method focuses primarily on peel colour, the Watershed method depends on the quality of foreground segmentation, and the CLAHE–Otsu method emphasises texture, intensity and gradient information. The Original RGB model serves as a common baseline for determining whether each proposed technique improves classification.

4.2.1 Overall Performance Comparison

Table 4.5 compares the final result of each proposed method with the Original RGB baseline. For the Watershed approach, the morphologically recovered result is used because it performed better than the initial Marker-Controlled Watershed result.

Table 4.9: Overall comparison of the three proposed methods

Method	Features	Test accuracy	Macro F1	Correct	Incorrect	Change from baseline
Original RGB baseline	–	83.00%	83.00%	167	33	–
Adaptive HSV and CIE L*a*b*	121	**92.00%**	**91.89%**	**184**	**16**	**+9.00%**
Watershed with boundary recovery	12,288	82.00%	82.04%	164	36	−1.00%
CLAHE, Otsu, LBP and HOG	351	82.50%	82.34%	165	35	−0.50%

The Adaptive HSV and CIE L*a*b* method achieved the highest accuracy at 92.00%. It exceeded the CLAHE and texture-based method by 9.50 percentage points and the refined Watershed method by 10.00 percentage points. It also reduced the number of errors from 33 in the baseline to 16, representing a 51.51% reduction:

Error reduction = (33-16) / 33 x 100% = 51.51%

The CLAHE and texture-based method ranked second with an accuracy of 82.50%. It correctly classified 165 images, five fewer than the adaptive colour method, but seven more than the RGB baseline. This indicates that contrast-enhanced texture and gradient information provide useful ripeness information, particularly for bananas with visible spots and surface deterioration.

The refined Watershed method achieved the lowest final accuracy at 82.00%. Although morphological boundary recovery improved the initial Watershed result from 78.50% to 82.00%, its performance

remained four percentage points below the baseline. Therefore, foreground segmentation alone did not guarantee improved classification. The usefulness of segmentation depended on whether the resulting mask preserved the colour and boundary information required by the SVM.

4.2.2 Interpretation of the Adaptive Colour Method

The Adaptive HSV and CIE L*a*b* method performed particularly well for unripe and ripe bananas. Its unripe recall increased from 0.66 in the baseline to 1.00, meaning that all 50 unripe test images were identified correctly. Its ripe precision also increased from 0.67 to 0.92.

This improvement suggests that separating hue, saturation, lightness and chromatic information made green and yellow peel characteristics easier for the SVM to distinguish. Foreground extraction also reduced the influence of backgrounds, plates, shadows, text and watermarks. The adaptive thresholds allowed feature selection to respond to the colour distribution of each image instead of applying one fixed threshold to every banana.

The method nevertheless performed less strongly for overripe bananas, whose recall was 0.82. Four overripe images were classified as ripe and another four as rotten. This reflects the transitional visual nature of the overripe class. Some overripe bananas remain predominantly yellow and resemble ripe samples, whereas others contain extensive dark spots and resemble rotten samples.

The adaptive method used only 121 features, substantially fewer than the 8,126 features used by the CLAHE method and the 12,288 features used by the Watershed method. Achieving the highest accuracy with the smallest feature vector suggests that the selected colour statistics and histograms were more directly related to the ripeness-classification objective. However, actual training and prediction times for all three methods would be required before making a definitive conclusion about computational efficiency.

4.2.3 Interpretation of the Watershed Method

The initial Marker-Controlled Watershed model achieved 79.50% accuracy. Morphological boundary recovery increased this result to 80.00%, showing that mask completeness influenced classification performance. The recovered boundary preserved a larger part of the banana and improved the correct predictions for unripe bananas from 35 to 46 and for overripe bananas from 38 to 42.

However, the recovery process did not improve every class. Correct ripe predictions decreased from 39 to 34, while correct rotten predictions decreased from 45 to 42. In particular, 15 ripe images were classified as rotten after boundary recovery. This suggests that the recovery operation sometimes reintroduced dark background regions, shadows or other pixels that resembled rotten surface areas.

The result demonstrates an important limitation of segmentation-based classification: retaining more pixels is not necessarily beneficial if the additional pixels do not belong to the fruit. Conversely, an incomplete mask may remove peel boundaries, brown spots or stem regions containing useful information. Watershed is therefore highly dependent on marker placement, gradient quality and post-processing parameters.

The use of 12,288 pixel-based features may also have contributed to the lower performance. A high-dimensional representation can contain redundant background, shape and zero-valued pixels. These variables may make it more difficult for the SVM to identify the most relevant ripeness characteristics, particularly when only 1,000 training images are available.

4.2.4 Interpretation of the CLAHE and Texture Method

The proposed CLAHE–Otsu–LBP–HOG method achieved a test accuracy of 82.50% and a macro F1-score of 82.34%. The model correctly classified 165 out of 200 test images, while 35 images were incorrectly classified. These results indicate that contrast enhancement, threshold-based structural information and handcrafted texture features provide useful information for banana-condition classification, but the resulting representation is less effective overall than the colour-based Adaptive HSV and CIE L*a*b* approach.

The Rotten class produced the strongest performance, achieving a precision of 0.8750, recall of 0.9800 and F1-score of 0.9245. The model correctly identified 49 out of 50 rotten images. The strong recall suggests that the texture and intensity characteristics enhanced by CLAHE were effective at recognising severe deterioration. Rotten bananas generally exhibit stronger dark regions and more obvious surface irregularities, which can be represented using grayscale intensity and local texture features.

The Overripe class also performed well, achieving a precision of 0.9149, recall of 0.8600 and F1-score of 0.8866. This indicates that LBP, HOG and grayscale statistical features were able to capture structural and surface variations associated with overripe bananas. However, some overripe samples remained difficult to distinguish from neighbouring stages because their appearance can overlap with both ripe and rotten bananas.

The weaker results occurred for the Unripe and Ripe classes. Unripe achieved an F1-score of 0.7500, while Ripe obtained 0.7327. Their recalls were 0.72 and 0.74 respectively. This suggests that grayscale texture and structural information alone are less effective for distinguishing early ripeness stages. The main visual difference between unripe and ripe bananas is often their green-to-yellow colour transition, which is not directly represented by the grayscale-based feature vector.

The current method also achieved a 99.30% training accuracy, substantially higher than its 82.50% test accuracy. The resulting 16.80 percentage-point difference indicates a degree of overfitting or limited generalisation. This may be related to the relatively small number of training images and the visual variation within the dataset.

4.2.5 Class-Level Comparison

Table 4.10: Comparison of Three Techniques

Class	Adaptive HSV–Lab F1	Watershed recovery F1	CLAHE–texture F1	Best method
Unripe	**0.9709**	0.93	0.85	Adaptive HSV–Lab
Ripe	**0.9200**	0.75	0.83	Adaptive HSV–Lab
Overripe	0.8542	0.85	**0.95**	CLAHE–texture
Rotten	0.9307	0.76	**0.95**	CLAHE–texture

The adaptive colour method was most effective for unripe and ripe bananas, where colour transition was the dominant distinction. In contrast, the CLAHE and texture method was strongest for overripe and rotten bananas, where surface defects and dark patterns became more prominent. The Watershed method did not produce the best F1-score for any class, although its unripe performance improved substantially after morphological boundary recovery.

These findings indicate that colour and texture provide complementary ripeness information. Colour is more useful for early ripeness stages, while texture becomes increasingly important during deterioration. A future hybrid model could combine the compact adaptive colour features with a carefully selected subset of LBP or HOG features.

4.2.6 Selection of the Best Method

Based on overall accuracy, macro F1-score, number of correct predictions and feature dimensionality, the **Adaptive HSV and CIE L*a*b* method is the best-performing method** in this experiment. It achieved:

- The highest test accuracy: 92.00%
- The highest macro F1-score: 91.89%
- The largest improvement over the baseline: 6.00 percentage points
- The fewest errors: 16 out of 200
- The smallest feature vector: 121 features
- Perfect recall for the unripe class
- Strong precision and recall for the ripe and rotten classes

Its cross-validation accuracy of 92.99% was also close to its test accuracy of 92.00%, suggesting consistent performance on the held-out test data.

Nevertheless, this conclusion should be expressed as the **best observed result on the current dataset**, rather than proof of universal superiority. The difference between the adaptive and CLAHE methods represents only five test images. No statistical significance test was performed to determine whether this difference would remain stable under different train-test splits.

Furthermore, the dataset contains varying backgrounds, watermarks and image sources. These factors may introduce dataset-specific information that does not represent banana ripeness, demonstrating that GrabCut can still encounter difficulties when the fruit and background have similar colours or when the scene is highly complex.

Repeated cross-validation, paired significance testing such as McNemar's test, and evaluation on an external banana dataset would provide stronger evidence of generalization. Within the current experimental conditions, however, the adaptive HSV and CIE L*a*b* pipeline provides the strongest balance between accuracy, class-level performance and compact feature representation.

Conclusion

5.1 Achievements

- **Objective 1 Status: Fulfilled** – Three banana-ripeness classification pipelines were successfully developed and implemented: Adaptive HSV and CIE L*a*b* feature extraction, Marker-Controlled Watershed with Morphological Boundary Recovery, and CLAHE–Otsu contrast enhancement with LBP and HOG texture extraction. Each pipeline generated a numerical feature representation that was classified using an RBF-SVM.
- **Objective 2 Status: Partially Fulfilled** – Only one of the three implemented techniques met the targeted accuracy threshold of 85%. The Adaptive HSV and CIE L*a*b* method achieved 92.00% accuracy. However, both the CLAHE and texture-based method (82.50%) and the Watershed method with Morphological Boundary Recovery (82.00%) fell short of the targeted accuracy. Therefore, the objective was partially fulfilled as only a single technique met the performance requirement.
- **Objective 3 Status: Fulfilled** – The three techniques were compared using test accuracy, macro F1-score, number of correct predictions, class-level performance and feature dimensionality. The comparison showed that the Adaptive HSV and CIE L*a*b* method achieved the highest overall performance, with 92.00% accuracy, a 91.89% macro F1-score and 184 correct predictions out of 200 test images. It outperformed the CLAHE and texture-based method, which achieved 82.50% accuracy, and the refined Watershed method, which achieved 82.00% accuracy.
- Project Code Repository: https://github.com/Eugene-panda/imageProcessing
- Streamlit: https://imageprocessing-9eu2mhevku9l56yxsngnjn.streamlit.app/

5.2 Limitations and Future Works

Although the project successfully implemented and evaluated three banana-ripeness classification techniques, several limitations were identified. These limitations concern the dataset, foreground segmentation, feature representations, classification process and experimental evaluation.

5.2.1 Dataset Limitations

The dataset contained 1,200 images divided into four ripeness classes. Although the class distribution was balanced, the dataset was relatively small and contained images collected from different online sources. Consequently, variations in backgrounds, lighting, camera angles, image resolutions and watermarks were present.

Some backgrounds or watermarks may also be associated with particular classes. Therefore, a classifier trained using the original images could learn dataset-specific background information instead of relying entirely on banana characteristics. This limits confidence that the reported accuracy would be maintained for new images captured in supermarkets, farms or household environments.

The dataset also did not clearly represent differences among banana varieties. Different varieties may have different shapes, peel colours and deterioration patterns. Thus, the trained model may not generalise equally well to banana varieties that were not sufficiently represented during training.

5.2.2 Foreground-Segmentation Limitations

The foreground-segmentation pipeline successfully generated masks for all 1,200 images, and no image was excluded because of an empty or excessively small mask. This was achieved through the combination of adaptive HSV–CIELAB seed initialization, six-iteration GrabCut, rectangle initialization, central-position recovery, adaptive border-colour recovery, morphological cleaning and connected-component filtering.

Nevertheless, successful mask generation does not necessarily indicate perfect foreground segmentation. Visual inspection showed that some masks retained parts of plates, shadows, logos, banners or background regions. In other cases, small portions of the banana boundary, stem or dark peel were removed. Dark rotten bananas remained particularly challenging because their peel colour could be similar to a black background. Aggressive removal of dark-background artifacts could also remove genuine dark banana regions containing important deterioration information.

The segmentation pipeline uses assumptions derived from the current dataset, including that the banana is normally positioned near the image centre and that image-border pixels usually represent the background. These assumptions may not hold when a banana touches the image boundary, appears far from the centre or occupies most of the image. Likewise, the manually selected component-size, border-width and morphological parameters may not generalize to images captured under different conditions.

The Watershed method was also sensitive to foreground-marker generation and boundary quality. Morphological Boundary Recovery improved its accuracy from 79.50% to 80.00%, but the additional

recovered pixels increased confusion between some ripe and rotten samples. This demonstrates that recovering a larger foreground region does not necessarily improve classification when the recovered region includes shadows or background noise.

No manually annotated ground-truth masks were available. Therefore, segmentation quality was mainly assessed through visual inspection and its effect on classification accuracy rather than direct quantitative segmentation metrics. Future evaluation should use manually annotated fruit masks to calculate Intersection over Union, Dice coefficient, segmentation precision and segmentation recall.

5.2.3 Feature-Extraction Limitations

The Adaptive HSV and CIE L*a*b* technique used several manually selected threshold ranges. For example, the adaptive saturation and chroma values were restricted using predefined clipping boundaries. These values performed well on the current dataset but may not be optimal under different lighting conditions, banana varieties or camera settings.

The proposed technique also experienced difficulty with the overripe class, which obtained a recall of 0.82. Overripe bananas form a transitional category because some samples resemble ripe bananas, while others contain dark areas similar to rotten bananas. Colour features alone may therefore be insufficient for consistently representing this class.

The CLAHE pipeline used 8,126 features, while the Watershed representation contained 12,288 features. These high-dimensional representations may contain redundant or irrelevant variables. Compared with the 1,000 training images, the large number of features could increase computational requirements and the risk of overfitting.

5.2.4 Future Work

Future work should first expand the dataset with images collected under controlled and real-world conditions. The expanded dataset should include different banana varieties, lighting conditions, camera devices, orientations, backgrounds and levels of surface damage. Duplicate images and watermarks should be removed, and the data should be divided carefully to prevent visually similar images from appearing in both training and testing sets.

Manual banana-mask annotations should also be created for a representative subset of the dataset. These annotations would allow GrabCut and Watershed segmentation to be evaluated quantitatively using Dice coefficient and Intersection over Union. GrabCut could be improved using automatic object detection, adaptive rectangle initialization and stronger foreground seeds. Semantic-segmentation models such as U-Net or DeepLab could also be investigated and compared with GrabCut.

A hybrid feature representation could combine the 121 adaptive colour features with selected LBP or HOG texture features. This may retain the adaptive model’s strong performance for unripe and ripe bananas while incorporating the texture method’s strong performance for overripe and rotten bananas.

Feature-selection techniques such as recursive feature elimination, permutation importance or principal component analysis could be applied to remove redundant features and control dimensionality.

The manually defined adaptive-threshold boundaries should be optimised using training data and nested cross-validation. Illumination normalization, colour calibration and data augmentation could also be introduced to improve robustness to brightness, contrast and camera variations.

Because the largest remaining difficulty concerns the overripe class, a hierarchical classifier could first distinguish early-stage bananas from deteriorated bananas and then separate ripe from unripe or overripe from rotten. An ordinal-classification approach could also be explored because the four labels represent an ordered ripeness progression.

Finally, future evaluation should include repeated stratified cross-validation, confidence intervals and paired statistical testing such as McNemar's test. Testing on an independent external dataset would provide stronger evidence that the selected method generalises beyond the current images. For practical deployment, all methods should also be compared using training time, prediction latency, memory usage and performance on mobile or low-resource devices.

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Appendix A

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY
Group Contract Form

Faculty : Faculty of Computing and Information Technology (FOCS)
Programme : Bachelor in Software Engineering (Honours)
Year of Study : 3
Semester : 1
Course Title : Image Processing
Course Code : BMDS2133
Group Assignment / Project Title : Comparative Analysis of Image Processing Techniques for Banana Ripeness Classification Using Support Vector Machines
Name of Tutor : Ms. Nurul Akmar binti Azman
Tutorial Group : 5

Student Name and ID	Role(s)/Task(s) Assigned	Signature and Date
Sia Jin Han 25WMR10042	Preprocessing / Adaptive HSV + CIELAB	 08/09/2026
Ho Jun Jie 25WMR09948	Preprocessing / Marker-Controlled Watershed + Morphological Boundary Recovery	 08/09/2026
Seow Kim Hui 25WMR10038	Preprocessing / Contrast Enhancement and Texture Analysis (CLAHE & Thresholding)	 08/09/2026

Appendix B

AI Usage Disclosure Form

Student Name : Sia Jin Han

Programme : Bachelor in Software Engineering (Honours)

Course : BMDS2133 Image Processing

Assessment Title : Comparative Analysis of Image Processing Techniques for Banana Ripeness Classification Using Support Vector Machines

1. AI Tool(s) Used (Check that all that apply)

- ☐ None (I did not use any AI tools for this assignment)
- ☑ ChatGPT (Version: 5.6 sol)
- ☐ Deepseek
- ☐ Gemini
- ☐ Other: _____

2. Nature of Assistance (Check all that apply)

- ☑ **Brainstorming:** Generating ideas, topics, or outlines.
- ☑ **Research:** Summarising long articles or finding concepts.
- ☑ **Editing:** Checking grammar, spelling, or sentence structure.
- ☑ **Coding/Math:** Debugging code or explaining a formula.
- ☑ **Creation:** Generating images, data, or draft text.

3. Disclosure Statement

Provide a brief explanation of how AI tools were utilised in completing the task and describe the measures taken to ensure that the final submission reflects your own independent work.

Example:
"I used ChatGPT to structure my initial ideas into an outline. Subsequently, I composed the full paragraphs independently and cross-checked the referenced dates against my textbook to ensure accuracy."

Your Statement:

I used ChatGPT to assist with structuring the report, explaining image-processing concepts, developing and debugging Python code, and improving the clarity of selected sections. I independently executed the code in Google Colab, reviewed the generated masks and classification results, verified the reported

accuracy and metrics, and checked the cited sources. I evaluated and revised all AI-assisted content to ensure that the final submission accurately represents the implemented system, experimental findings, and my own understanding of the project.

4. Verification & Integrity

- ☑ I have reviewed and verified the accuracy of all facts, data, and citations generated with the assistance of AI tools.
- ☑ I have properly acknowledged and cited any AI-generated text, ideas, or content in accordance with my instructor's requirements.
- ☑ I confirm that the reasoning, interpretation, and analysis presented in the final submission are entirely my own.

Student Signature : Sio

Date : **08/09/2026**

AMM 5.3.2026/SMC 1.4.2026

AI Usage Disclosure Form

Student Name : Ho Jun Jie

Programme : Bachelor in Software Engineering (Honours)

Course : BMDS2133 Image Processing

Assessment Title : Comparative Analysis of Image Processing Techniques for Banana Ripeness Classification Using Support Vector Machines

1. AI Tool(s) Used (Check that all that apply)

- ☐ None (I did not use any AI tools for this assignment)
- ☑ ChatGPT (Version: 5.6 sol)
- ☐ Deepseek
- ☐ Gemini
- ☐ Other: _____

2. Nature of Assistance (Check all that apply)

- ☑ **Brainstorming:** Generating ideas, topics, or outlines.
- ☑ **Research:** Summarising long articles or finding concepts.
- ☑ **Editing:** Checking grammar, spelling, or sentence structure.
- ☑ **Coding/Math:** Debugging code or explaining a formula.
- ☑ **Creation:** Generating images, data, or draft text.

3. Disclosure Statement

Provide a brief explanation of how AI tools were utilised in completing the task and describe the measures taken to ensure that the final submission reflects your own independent work.

Example:
"I used ChatGPT to structure my initial ideas into an outline. Subsequently, I composed the full paragraphs independently and cross-checked the referenced dates against my textbook to ensure accuracy."

Your Statement:

I used ChatGPT mainly to assist with organising the report structure, improving the clarity of technical explanations, debugging selected Python code, and refining descriptions of the image-processing methods used in the project. I independently implemented and tested the assigned Marker-Controlled Watershed and Morphological Boundary Recovery pipeline, reviewed the generated outputs, and verified the experimental results against the actual notebook execution. All final content, figures,

parameter choices, interpretations, and conclusions were checked and adjusted to ensure that the submitted work accurately reflects the methods and results produced by our group.

4. Verification & Integrity

- ☑ I have reviewed and verified the accuracy of all facts, data, and citations generated with the assistance of AI tools.
- ☑ I have properly acknowledged and cited any AI-generated text, ideas, or content in accordance with my instructor's requirements.
- ☑ I confirm that the reasoning, interpretation, and analysis presented in the final submission are entirely my own.

Student Signature : 

Date : **08/09/2026**

AMM 5.3.2026/SMC 1.4.2026

AI Usage Disclosure Form

Student Name : **Seow Kim Hui**

Programme : **Bachelor in Software Engineering (Honours)**

Course : **BMDS2133 Image Processing**

Assessment Title : **Comparative Analysis of Image Processing Techniques for Banana Ripeness Classification Using Support Vector Machines**

1. AI Tool(s) Used (Check that all that apply)

- ☐ None (I did not use any AI tools for this assignment)
- ☑ ChatGPT (Version: 5.6 sol)
- ☐ Deepseek
- ☑ Gemini
- ☐ Other: _____

2. Nature of Assistance (Check all that apply)

- ☐ **Brainstorming:** Generating ideas, topics, or outlines.
- ☑ **Research:** Summarising long articles or finding concepts.
- ☑ **Editing:** Checking grammar, spelling, or sentence structure.
- ☑ **Coding/Math:** Debugging code or explaining a formula.
- ☐ **Creation:** Generating images, data, or draft text.

3. Disclosure Statement

Provide a brief explanation of how AI tools were utilised in completing the task and describe the measures taken to ensure that the final submission reflects your own independent work.

Example:
"I used ChatGPT to structure my initial ideas into an outline. Subsequently, I composed the full paragraphs independently and cross-checked the referenced dates against my textbook to ensure accuracy."

Your Statement:

I used ChatGPT and Gemini to brainstorm the project structure, draft sections of the literature review, and generate baseline Python code for the image processing algorithms. To ensure the final submission reflects my independent work, I manually tested and modified the code to fit our specific fruit dataset, adjusted the algorithm parameters (such as the CLAHE clip limits), and cross-checked all AI-provided academic citations against their original published sources to verify their accuracy.

4. Verification & Integrity

- ☑ I have reviewed and verified the accuracy of all facts, data, and citations generated with the assistance of AI tools.
- ☑ I have properly acknowledged and cited any AI-generated text, ideas, or content in accordance with my instructor's requirements.
- ☑ I confirm that the reasoning, interpretation, and analysis presented in the final submission are entirely my own.

Student Signature : 

Date : **08/09/2026**

AMM 5.3.2026/SMC 1.4.2026